Check for updates

ACM DIGITAL LIBRARY

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3736750

PDF Download
3736750.pdf
18 February 2026
Total Citations: 7
Total Downloads: 1275

Published: 11 July 2025
Online AM: 27 May 2025
Accepted: 18 April 2025
Revised: 06 February 2025
Received: 04 December 2021

Citation in BibTeX format

SURVEY

Sentiment Diffusion in Online Social Networks: A Survey from the Computational Perspective

HAN XU, Huazhong University of Science and Technology, Wuhan, Hubei, China

MINGHUA XU, Huazhong University of Science and Technology, Wuhan, Hubei, China

XIANJUN DENG, Huazhong University of Science and Technology, Wuhan, Hubei, China

BANG WANG, Huazhong University of Science and Technology, Wuhan, Hubei, China

Open Access Support provided by:

Huazhong University of Science and Technology

ACM Computing Surveys, Volume 57, Issue 12 (December 2025)
https://doi.org/10.1145/3736750
EISSN: 1557-7341

Sentiment Diffusion in Online Social Networks: A Survey from the Computational Perspective

HAN XU, School of Journalism and Information Communication, Huazhong University of Science and Technology, Wuhan, China
MINGHUA XU, School of Journalism and Information Communication, Huazhong University of Science and Technology, Wuhan, China
XIANJUN DENG, School of Cyber Science and Engineering, Huazhong University of Science and Technology, Wuhan, China
BANG WANG, School of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan, China

With the development of mobile technologies, users can easily access Online social networks (OSNs), consequently, massive contents including personal experiences, observations, or opinions are generated online. These contents are being shared and exchanged in OSNs, which have a significant influence on the minds of people toward politics, societies, economics, and so on. In this case, sentiment diffusion which focuses on how the process of information diffusion in OSNs is affected by sentiments has become an important issue. In this survey, we conduct a comprehensive review of the problem. Specifically, we first present the definition and classification of sentiment, introduce the most advanced computational sentiment analysis techniques, list their applications, and disclose available sentiment analysis resources. Then the main OSN components, functionalities, and structural features applied in modeling social networks are analyzed and summarized. Next, a thorough overview of the information diffusion technologies in OSNs, including graph and non-graph models is made. At last, a systematic and in-depth overview of the current state and issues of research on sentiment diffusion in OSNs is provided. This survey provides the necessary knowledge and new insights for relevant researchers to better understand the research state, remaining challenges, and future directions in this field.

CCS Concepts: • **Human-centered computing → Collaborative and social computing theory, concepts and paradigms**; • **General and reference → Surveys and overviews;**

Additional Key Words and Phrases: Sentiment diffusion, information diffusion, sentiment analysis, online social networks

This work was supported by the Major Project of the National Social Science Foundation of China (22&ZD317), the SMP-IDATA Chenxing Youth Fund (SMP2023-iData-013), the Open Projects Program of State Key Laboratory of Multimodal Artificial Intelligence Systems (MAIS-2023-18), and the Fundamental Research Funds for the Central Universities, JLU.
Authors' Contact Information: Han Xu, School of Journalism and Information Communication, Huazhong University of Science and Technology, Wuhan, Hubei, China; e-mail: xuh@hust.edu.cn; Minghua Xu, School of Journalism and Information Communication, Huazhong University of Science and Technology, Wuhan, Hubei, China; e-mail: xuminghua@hust.edu.cn; Xianjun Deng, School of Cyber Science and Engineering, Huazhong University of Science and Technology, Wuhan, Hubei, China; e-mail: dengxj615@hust.edu.cn; Bang Wang (Corresponding author), School of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan, Hubei, China; e-mail: wangbang@hust.edu.cn.
Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM 0360-0300/2025/07-ART298
https://doi.org/10.1145/3736750

ACM Reference Format:
Han Xu, Minghua Xu, Xianjun Deng, and Bang Wang. 2025. Sentiment Diffusion in Online Social Networks: A Survey from the Computational Perspective. *ACM Comput. Surv.* 57, 12, Article 298 (July 2025), 35 pages. https://doi.org/10.1145/3736750

1 Introduction

Online social networks (**OSNs**), such as Facebook, TikTok, X (formerly known asTwitter), Instagram, Weibo, and WeChat, are becoming an integral part of our lives and have already attracted a huge number of users during the past decade as mobile technologies evolved. For instance, as of March 2025, Facebook, TikTok, and X had 3.08 billion, 1.83 billion and 0.611 billion **monthly active users** (**MAUs**), respectively, and the number is still growing. The prevalence of OSNs has not only become indispensable information sources and entertainment channels in our daily lives but has also continuously changed how people recognize diverse physical worlds and how people establish various social relations. Given the short history of starting in 1997 marked by the launch of SixDegrees.com website, they have overwhelmingly shifted the paradigm of people communication, business operation, and political ecoenvironment [32, 52]. Apparently, OSNs have been constantly converging the physical and cyber social worlds, or so-called *cyber-physical convergence* [32].

For the fast penetration of OSNs, massive contents, including personal experiences, observations, or opinions are generated online by the users. These contents are being shared and exchanged in OSNs day by day, which has a significant influence on the minds of people toward anything like politics, societies, and economics. For example, if someone's friends have shown positive comments against a certain item in the OSNs, such as the COVID-19 vaccine, he or she may be influenced by those online contents and have a similar feeling; People usually need to take into account the others' comments, before purchasing a product, watching a film, and selecting restaurants. In this case, sentiment diffusion, which focuses on analyzing how the process of information diffusion in OSNs is affected by sentiments, has become a hot topic and attracted increasing attention from many research communities [171].

From the computational perspective, the problem about **sentiment diffusion in OSNs** is nontrivial and poses a set of unique challenges: (1) How to simultaneously capture and analyze the content and the structure information of a social network (**Social Network Analysis**)? (2) How many emotions are there in total? How to identify the user's sentiment hidden in his/her social context **Text Sentiment Analysis** (**TSA**)? (3) What kind of information or sentiment is likely to be diffused easily and widely? How many users it could reach? Which paths it could traverse most likely and which users could play what role during its diffusion (**Information and sentiment diffusion modeling**)? Obviously, it is a field that encompasses a plethora of techniques such as sentiment analysis, complex network analysis, and information diffusion.

Currently, there exist many surveys in the field of social networks [52, 115, 206], sentiment analysis [28, 39, 59, 112, 192], and information diffusion [65, 83, 101]. However, to the best of our knowledge, there are no existing articles that have reviewed the sentiment diffusion problem in OSNs. This motivates the need for a different and new perspective on the problem.

In this survey, we try to systematically review the current status, challenges, and future directions of this problem from the computational perspective. We conducted a comprehensive literature search on multiple databases, such as Web of Science, Google Scholar, IEEE Xplore, ACM digital library, and Elsevier ScienceDirect. During the search, we mainly used keywords such as "sentiment diffusion", "information diffusion'," and "social network dynamics" to search literature related to the topic. To avoid omission, we also searched for related terms such as "sentiment

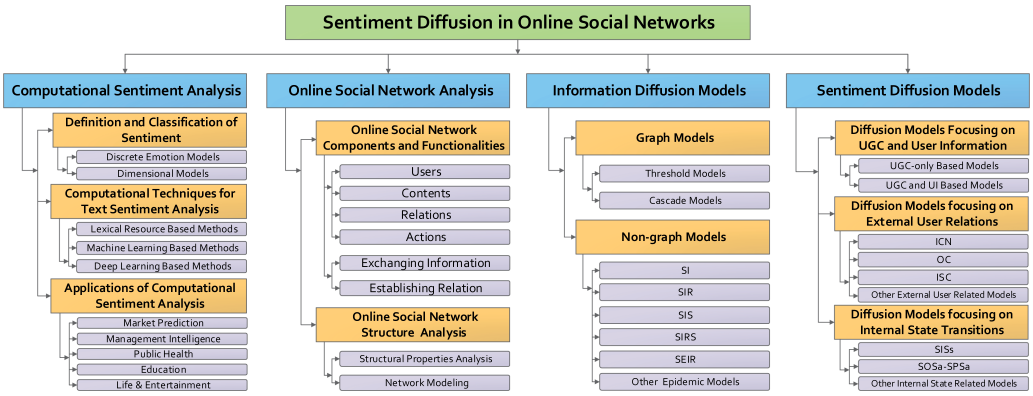


Fig. 1. Overview of recent advances on sentiment diffusion in OSNs included in this survey.

analysis", "threshold model", "cascade model", "SIR", and "SIS". Combined with work related to sentiment analysis, complex network dynamics and information diffusion, we believe that this review can provide a clear overview of studies researching sentiment diffusion in OSNs as a whole. Figure 1 gives an overview of this survey paper. In summary, we make the following contributions in this research:

- We comprehensively present the classification of sentiment, including basic emotion models and dimensional models.
- We introduce some basic components and the most advanced approaches in TSA, which are important for sentiment diffusion in OSNs. Available sentiment analysis resources are also disclosed.
- OSNs components and functionalities, structural properties analysis, and network modeling are summarized and discussed.
- Information diffusion technologies in OSNs, including graph models and non-graph models are presented.
- Sentiment diffusion models focusing on UGC and user information, external user relations, and internal state transitions are discussed. Moreover, future directions of the problem are provided from a computational perspective.

The rest of this survey article is organized as follows. In Section 2, we present the definition and classification of sentiment, then introduce three kinds of computational techniques of TSA and their applications in different fields. Section 3 reviews the basics of OSN components, functionalities, structures, and modeling methods. Section 4 reviews technologies and models for information diffusion in OSNs. Sentiment diffusion models focusing on UGCs, user information, external user relations, and internal state transitions are discussed in Section 5. In Section 6, we offer future research directions. Finally, the conclusion are presented in Section 7.

2 Computational Sentiment Analysis

2.1 Definition and Classification of Sentiment

Sentiment is a very complicated psychological concept, so far, psychologists have not reached a consensus on the concept and connotation of sentiment. For example, some researchers believe that sentiments are hardwired physiological patterns [133], while others believe that sentiments are the result of social constructions [8]. According to the *Merriam-Webster's Online Dictionary*,[1]

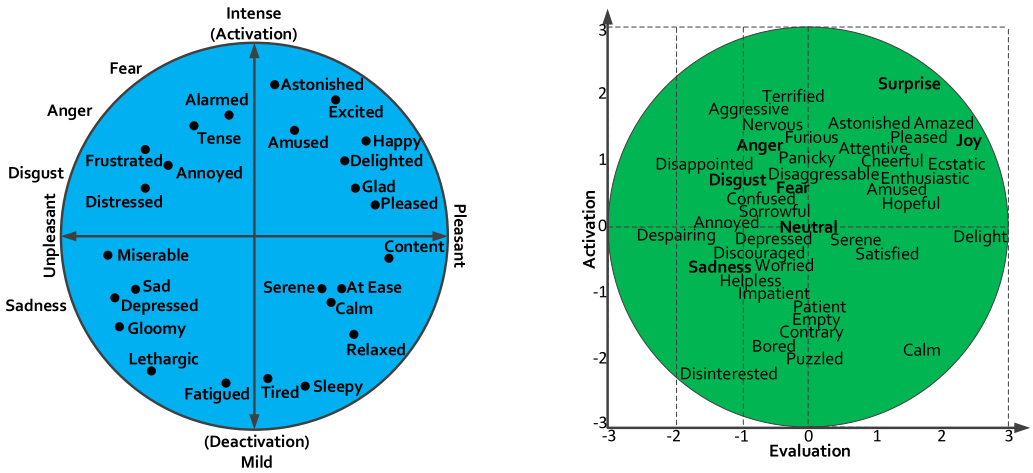
[1] https://www.merriam-webster.com/dictionary/sentiment

sentiment can be described as the emotion (the affective aspect of consciousness) aroused by or the opinion (a specific view or notion) towards some entity. Here, an entity could be some product, service, event, video, picture, article, and so on. Some other synonyms of sentiment include chord, affect, feeling, and mood. Notice that in this article, the subject of sentiment is a human, other than a robot. People can express their sentiments via either explicit physiological change in the body with visible cues, such as crying and laughing, or implicit psychological change in the mind with invisible cues, such as sadness and happiness.

Capturing and characterizing human sentiment is of great importance in many applications. For example, if a social robot can detect human affect from facial expression, body language, voice tone, and/or physiological signals, it can provide better services to fit into people's daily lives. Diverse algorithms and systems have been designed to automatically detect the affective states based on human physiological change [127, 151, 168]. Yet in this article, we are mainly concerned on another important expression of sentiments that are embedded in textual materials. For example, a smartphone buyer posted a comment on a digital product website, saying "*I like the newly bought smartphone. It has a large screen and a long battery life*". we can infer that some positive sentiment of this buyer to the product.

Modern research suggests that sentiments/emotions involve multiple different components, such as subjective experience, cognitive processes, expressive behavior, psychophysiological changes, and instrumental behavior [152]. As a result, basic emotion theory and dimensional theory have emerged as two representative and controversial theories in the study of emotion structure [150]. Basic emotion theory, also known as discrete emotion theory, holds that the basic units of emotion are discrete in nature, such as happiness and sadness. These basic emotions have consistent and specific psychological, physiological, and neurological correlations [91]. Dimensional theory says that emotion is a highly correlated continuum and a relatively fuzzy state, and it is difficult to distinguish various specific emotions. Various emotions are highly correlated in several basic dimensions, so different dimensions or cores of emotions should be grasped to explain them [181].

2.1.1 Basic Emotion Models. Due to their simplicity and intuitiveness, discrete emotion models have been widely accepted to analyze what kind of sentiment has been embedded in a piece of text. The discrete emotion label usually consists of a limited number of basic emotions that are independent of each other [130]. Theories about the basic emotion category originated from ancient Greece and China. The ancient Greek philosopher Aristotle divided emotions into *desire, anger, terror, joy,* and *pity*. The ancient Chinese encyclopedia "The Book of Rites" identifies seven "emotions of human": *joy, anger, sadness, fear, love, disliking,* and *liking*. Current discrete emotion research started with Charles Darwin in 1872, after he compiled the famous book "On the Origin of Species". In his work, Darwin treated the emotions as separate discrete entities, such as *anger, fear, disgust, anger, surprise,* and *disgust* [38]. Based on bodily involvement, William James [79] proposed four basic emotions: *fear, grief, love*, and *rage* in 1890. Subsequently, William McDougall [117] distinguished certain emotions as primary emotions, namely, *anger, disgust, elation, fear, subjection, tender-emotion,* and *wonder*. In modern research, even though many researchers have accepted the theory of basic discrete emotions, there is no consensus on the categories of emotions [39]. Following Darwin's view of emotion and based on the study of facial expressions, Paul Ekman proposed six widely recognized basic emotions, sometimes referred to as "The Big Six", namely, *anger, disgust, fear, happiness, sadness,* and *surprise* [49]. In the 1990s, Ekman expanded the basic emotions list, and the newly included emotions are: *amusement, contempt, contentment, embarrassment, excitement, guilt, pride in achievement, relief, satisfaction, sensory pleasure*, and *shame* [50]. Psychologist Silvan Tomkins developed the Affect Theory [170]



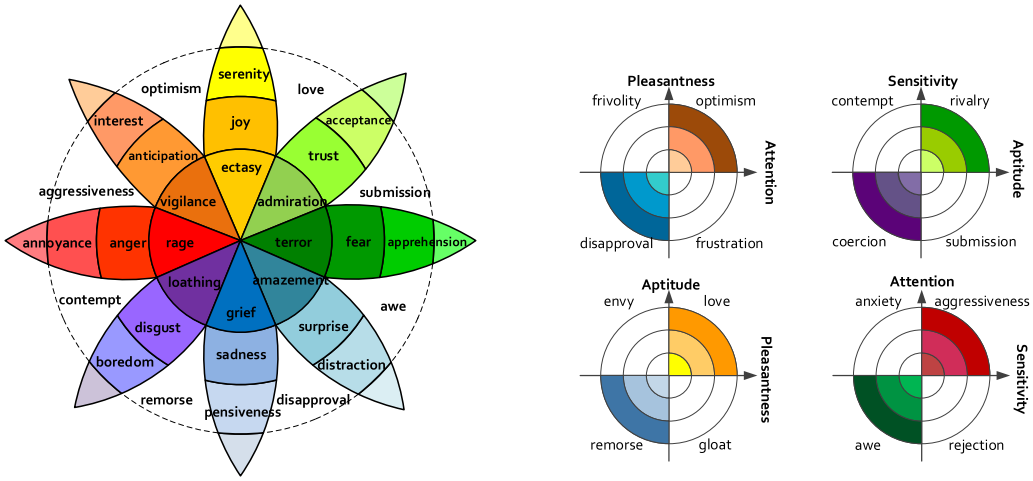
(a) (b)

Fig. 2. (a) The circumplex model of emotion by Russell [148]. (b) The bi-dimensional representation of emotions by Whissell [185].

from the 1960s until his death in 1992. According to the Affect Theory, there are nine basic affects: The positive ones are *Interest-Excitement, Enjoyment-Joy*; the Neutral one is *Startle-Surprise*; and the Negative ones are *Distress-Anguish, Anger-Rage, Fear-Terror, Shame-Humiliation, Disgust, and Dissmell.* Carroll E. Izard proposed **Differential Emotions Theory** (**DET**) and initially suggested that there are 10 basic emotions [76]. In addition to the big six, there are also *interest, shame, guilt,* and *contempt.* Nico Frijda [54] discussed the motivational and neurophysiological preconditions for emotions and proposed that there are six basic emotions: *desire, happiness, interest, surprise, wonder,* and *sorrow.* Richard Lazarus and Bernice Lazarus [96] expanded the list to 15 emotions: *aesthetic experience, anger, anxiety, compassion, depression, envy, fright, gratitude, guilt, happiness, hope, jealousy, love, pride, relief, sadness*, and *shame.* Some studies believe that in Ekman's study, the facial expression of disgust is similar to the facial expression of anger, as well as the facial expression of surprise and fear. Therefore, they believe that emotions should be divided into *anger, fear, happiness,* and *sadness* [78]. Varghese and Thampi [172] likewise omitted *surprise* and *disgust* and added *excited* and *neuter.* Cowen and Keltner [34] determined 27 different emotional categories by studying thousands of people's responses to different videos, music, and various situations.

2.1.2 Dimensional Models. When recording and communicating different sentiments through language, emotion keywords, and their semantic meanings in contexts are often used to describe the spectrum of basic emotions. Considering that some basic emotions are related to each other, some scholars began to express emotions in different dimensions and thus came into being the dimensional emotion model, which uses two-dimensional, three-dimensional, and even four-dimensional system to express emotion space [118].

The *circumplex model* and its variants [13, 140, 148] distribute basic emotions onto a two-dimensional circular plane, with a vertical axis as *Arousal* and a horizontal axis as *Valence.* The center of the circle represents a neutral valence and a medium level of arousal. Generally, emotional valence in psychology describes the extent to which an individual is attracted or repelled towards an entity ranging from unpleasantness to pleasantness; While emotional arousal describes the level of physiological intensity of the sentiment evoked by some entity ranging from low to



(a) (b)

Fig. 3. (a) The Plutchik emotion model [139]. (b) The Hourglass compound emotions [26].

high. As illustrated by Figure 2(a), each emotion can find its location in such a circumplex model. For example, anger and sadness are considered as unpleasant emotions, while anger is often characterized by high arousal and sadness by low arousal. Similar two-dimensional emotional models using the *Valence*-dimensions and *Arousal*-dimensions include the Vector model [21] and the **Positive Activation–Negative Activation** (**PANA**) model [182], which also adopt both valence and arousal as dimensions. Note that in the PANA model, the dimensions of *Valence* and *Arousal* lay at a 45-degree rotation over two axes of positive affect and negative affect. Whissell [185] also considered emotions as a continuous 2D space whose dimensions are *evaluation* and *activation*, as shown in Figure 2(b).

As early as 1897, Wilhelm Max Wundt [188] first proposed the three-dimensional emotional model, which divides emotions into three dimensions: *pleasurable versus unpleasurable*, *arousing or subduing*, and *strain or relaxation*. In 1954, Harold Schlosberg [154] proposed three dimensions of emotion: *pleasantness–unpleasantness, attention–rejection,* and *level of activation*. Through factor analysis, Osgood found that emotional experience can be evaluated from the following three dimensions: *evaluation, potency,* and *activity*. Inspired by Osgood's work, Mehrabian and Russell proposed a three-dimensional ***Pleasure-Arousal-Dominance*** (**PAD**) emotion model. According to these three dimensions, emotions can be divided into eight categories [118, 148]. The Plutchik emotional model [139] divides emotions into eight primary groups, *joy, trust, fear, surprise, sadness, anticipation, anger,* and *disgust*, and each primary group has a polar opposite peer, such as joy being opposite of sadness, fear the opposite of anger, as shown in Figure 3(a). Furthermore, each primary emotion group can be described with different intensities, like *serenity, joy,* and *ecstasy* in the *joy* group with intensity from low to high, and different emotions can mix with one another to form other combinatory emotions, like *optimism* being combined by *joy* and *anticipation*.

Carroll E. Izard also proposed the four-dimensional theory of emotion, which holds that emotions have four dimensions: *pleasure, tension, excitement,* and *conviction* [77]. Cambria et al. [26] proposed the hourglass of emotion model with 24 detailed features based on four dimensions of *pleasure, attention, inclination,* and *sensitivity* and three polar strengths, as shown in Figure 3(b). Moreover, Fontain et al. [53] proposed a four-dimensional emotional space containing *valence, potential, activation*, and *unpredictability*.

Table 1. Summary of the Emotion Models

Type	Author	Year	Categories	Emotions / Dimensions
Discrete emotion models	Aristotle	335 BC	5	desire, anger, terror, joy, and pity
	Dai	30 BC	7	joy, anger, sadness, fear, love, disliking, and liking
	Darwin [38]	1872	6	anger, fear, disgust, anger, surprise, and disgust
	James [79]	1890	4	fear, grief, love, and rage
	McDougall	1920s	7	anger, disgust, elation, fear, subjection, tender-emotion, and wonder
	Ekman [49]	1960s	6	anger, disgust, fear, happiness, sadness, and surprise
	Tomkins [170]	1960s	9	interest-excitement, enjoyment-joy, startle-surprise, distress-anguish, anger-rage, fear-terror, shame-humiliation, disgust, and dissmell
	Izard [76]	1971	10	anger, contempt, disgust, distress, fear, guilt, interest, joy, shame, and surprise
	Frijda	1986	6	desire, happiness, interest, surprise, wonder, and sorrow
	Ekman [50]	1990s	11	amusement, contempt, contentment, embarrassment, excitement, guilt, pride in achievement, relief, satisfaction, sensory pleasure, and shame
	Lazarus [96]	1994	15	anger, fright, anxiety, guilt, shame, sadness, envy, jealousy, disgust, happiness, pride, relief, love, hope, and compassion
	Cowen [34]	2017	27	admiration, adoration, aesthetic appreciation, amusement, anxiety, awe, awkwardness, boredom, calmness, confusion, craving, disgust, empathetic pain, entrancement, envy, excitement, fear, horror, interest, joy, nostalgia, romance, sadness, satisfaction, sexual desire, sympathy, and triumph
	Varghese [172]	2018	6	anger, fear, happiness, sadness, excited, and neuter
Dimensional models	Russell [148]	1980	2	valence, arousal
	Watson [182]	1985	2	valence, arousal
	Bradley [21]	1992	2	valence, arousal
	Whissell [185]	1989	2	evaluation, activation
	Wundt [188]	1897	3	pleasant-unpleasant, tension-relaxation, and excitement-stillness
	Plutchik [139]	2001	3	intensity, similarity, and bipolarity
	Schlosberg [154]	1954	3	pleasantness–unpleasantness, attention–rejection, and level of activation
	Mehrabian [118]	1996	3	pleasure-displeasure, arousal-nonarousal, and dominance-submissiveness
	Izard [77]	1977	4	intensity, tension, pleasure, and complexity
	Fontain [53]	2007	4	pleasure, tension, excitement, and conviction
	Cambria [26]	2012	4	pleasure, attention, inclination, and sensitivity

2.1.3 Basic vs. Dimensional Models. The features of the two types of emotion representation theories[2] are summarized in Table 1. The psychological and biological theories on basic emotions and emotion dimensions are based on qualitative and quantitative subject tests based on visual and verbal cues. Although it has been critically argued that human emotion may be ethnographic- and cultural-dependent [149], these classifications, on the one hand, could provide universal dimensions for sentiment measurement recognizable to all peoples. On the other hand, basic emotions and dimensional emotion models have also paved the way for the computational analysis of TSA, which has become a hot interdisciplinary research field in the last decades [108].

Each of these two types of emotion theories has advantages and disadvantages. The basic emotion models provide a representation of emotions that can be understood and experienced by

[2]Another important stream in emotion theory is the Appraisal Theory, which emphasizes the process of emotions based on people's interpretations and reactions to their circumstances. Since most of the current computational techniques for TSA are based on the basic and dimensional sentiment models, we mainly provide a brief review of these two types of models in this article. For more references on the Appraisal Theory, interested readers may refer to [95, 153].

humans, but the types of emotions represented are limited, can only be described qualitatively, and do not fully reflect the complex and subtle psycho-emotional states of humans. Dimensional emotion models can theoretically represent all human emotional states, but they are usually not easily understood and experienced because they are represented as coordinate values in dimensional space.

2.2 Computational Techniques for Text Sentiment Analysis

TSA is to analyze whether a piece of text contains some emotion and if does, which type of emotion can be detected. Many computational techniques have been proposed to implement automatical TSA and some recent survey articles can be found in [28, 39, 59, 112, 192]. In this article, we do not intend to provide a comprehensive survey on computational TSA. Instead, we briefly introduce some basic components and approaches in this field. Note that different computational techniques can be basically divided into three categories, including lexical resource-based, machine learning-based, and deep learning based-methods, which are summarized in Table 2.

2.2.1 Lexical Resource Based Methods. The methods based on lexical resource refer to using the emotional polarity of emotional words provided by different lexical resources for sentiment classification and opinion mining applications. SentiWordNet is one of the earliest lexical resource for TSA, which is the result of automatically annotating all the synsets of WORDNET according to their degrees of positivity, negativity, and neutrality [51]. Based on SentiWordNet, some studies try to improve the accuracy of TSA or build a new sentiment lexical resource. For example, Siersdorfer et al. [158] presented an in-depth study of commenting and comment rating behavior on a sample of more than 6 million YouTube comments using SentiWordNet thesaurus. Balahur et al. [11] used SentiWordNet to extend their model EmotiNet, which turned out to be a semantic resource appropriate for capturing and storing the structure and the semantics of real events and predicting the emotional responses triggered by chains of actions. By blending the most frequently used techniques based on SentiWordNet and popular manually annotated sentiment lexicon in an ensemble learning framework, Gattir et al. [58] built SentiWords, a prior polarity lexicon of approximately 155,000 words, that has both high precision and high coverage.

Surely, there were many other outstanding sentiment lexical resources, such as **Multi-perspective Question Answering** (**MPQA**) [186], **Semantic Orientation CALculator** (**SO-CAL**) [166], and **Linguistic Inquiry and Word Count** (**LIWC**) [20] that can be used for different TSA tasks.

All the above corpora are created from discrete emotional models. On the other hand, some researchers tried to construct high-quality dimensional emotional corpora. For example, **Affective Norm for English Words** (**ANEW**) [22] and NRC_VAD [122] are both annotated with Valence and Arousal scores. SenticNet-6 [25] is automatically annotated based on the Hourglass of Emotions and is available in 40 different languages.

However, TSA with only the approach of sentiment lexical resources has limitations in vocabulary coverage and domain adaptation [5]. Current researches are trying to integrate sentiment lexical resources with other new techniques, such as machine learning and deep learning. The details will be introduced in a hybrid approach in the following sections.

2.2.2 Machine Learning Based Methods. Machine learning techniques have been widely deployed for text sentiment classification, from the document level to the sentence and word/phrase level. Basically, this type of method trains models to predict text sentiments through a given dataset. By adopting different machine learning technologies, such as **Naïve Bayes** (**NB**), **Random Forest** (**RF**), **Support Vector Machine** (**SVM**), **K-Nearest Neighbor** (**KNN**), and **Latent Dirichlet**

Table 2. Different Computational Techniques for TSA

Type	Method/Model	Lexicon/Corpus/Dataset	Advantages
Lexical Resource Based Methods	MPQA [186]	A lexicon of over 8,000 subjectivity clues	Phrase-level sentiment analysis
	SentiWordNet [51]	General Inquirer lexicon	Wide coverage, fine grain
	EmotiNet [11]	ISEAR corpus[3]	Representing and storing affective reaction to real-life contexts
	SO-CAL [166]	Epinions, Movie, Camera[4]	Good cross-domain performance
	SentiWords [58]	General Inquirer[5]	High precision and high coverage
Machine Learning Based Methods	R-JST [106]	MR[6], MDS[7]	Highly portable to other domains
	GARCH [187]	Dataset crawled from Sina finance web	Has a higher classification accuracy than that of the semantic approach
	DSLL [189]	X and Weibo datasets	Has a good classification performance and computational efficiency
	MME [103]	ISEAR, SemEval[8]	Competitive compared with various baselines
Deep Learning Based Methods	Word2Vec [120]	An internal Google dataset with one billion words	Powerful, simple, minimal computational complexity
	GloVe [136]	CoNLL-2003[9]	Outperforms other models on word analogy, word similarity, and named entity recognition tasks
	ELMo [137]	SQuAD[10]	Significantly improve the state of the art across six challenging NLP problems
	Emo2Vec [191]	ISEAR, WASSA[11]	Has competitive performances to state-of-the-art results on several tasks
	WDE-CNN [202]	User-product-review graph[12]	Effective and outperforms baseline methods
	HNN [107]	A large number of Sina Weibo texts	More effective than other methods on Sina Weibo
	+cell [142]	CR[13]	Beneficial to many downstream NLP tasks
	BLS [45]	CrowdFlower[14], DL4J2[15]	Has higher accuracies than LSTM with less training time
	DL-BGDL [164]	MOOC[16], IMDB-10[17], STT-5[18]	Has a good performance on four datasets, and needs fewer parameters compared with recent benchmarks

(Continued)

[3]https://www.unige.ch/cisa/research/materials-and-online-research/research-material/
[4]www.sfu.ca/~{}mtaboada/research/nsercproject.html
[5]www.wjh.harvard.edu/Finquirer/
[6]http://reviews.imdb.com/Reviews/
[7]http://www.cs.cornell.edu/people/pabo/movie-review-data/
[8]https://alt.qcri.org/semeval2014/index.php?id=tasks
[9]https://paperswithcode.com/dataset/conll-2003
[10]https://rajpurkar.github.io/SQuAD-explorer/
[11]https://workshop-wassa.github.io/
[12]https://www.dropbox.com/s/aji68llxmtcuu5l/data.zip
[13]https://www.kaggle.com/datasets/cynthiarempel/amazon-us-customer-reviews-dataset
[14]https://www.crowdflower.com/data-for-everyone/
[15]https://deeplearning4j.org/cn/opendata
[16]http://datastage.stanford.edu/StanfordMoocPosts
[17]http://datasets.imdbws.com
[18]http://nlp.stanford.edu/sentiment

Table 2. Different Computational Techniques for TSA

Type	Method/Model	Lexicon/Corpus/Dataset	Advantages
	Regional CNN-LSTM [176]	SST[19], EmoBank[20], FB[21], CVAT[22]	Outperforms lexicon-, regression-, CNN and other structured NN methods
	Transformer [173]	WMT[23]	Superior in quality, more parallelizable, less time to train
	TextGT [195]	SemEval, Rest16	Over the state-of-the-art approaches
	SRAR [205]	LDC[24], WMT14[25], WMT16[26]	Helps self-attention networks outperform existing approaches
	RGAT [10]	SemEval, ACL14, X[27], MAMS[28]	Significantly outperforms state-of-the-art syntax-based approaches
	BERT [42]	GLUE[29]	Conceptually simple and empirically powerful
	CS-KG [102]	IMDB, MR, SST-5	Achieves new state-of-the-art results

Allocation (**LDA**), many researchers have focused on studying the sentiment of text on OSNs. For example, Li et al. [103] introduced a **multi-label maximum entropy** (**MME**) model for user emotion classification over short text. Based on LDA, Lin et al. [106] proposes a novel probabilistic modeling framework called **Joint Sentiment-Topic** (**JST**) model which detects sentiment and topic simultaneously from text at document level. Their experimental results on the **movie review** (**MR**) and **multidomain sentiment** (**MDS**) datasets can prove that JST outperforms existing semi-supervised approaches in some of the datasets. Wu et al. [187] developed a novel sentiment ontology to conduct context-sensitive sentiment analysis of online opinion posts in stock markets, whose methodology integrates popular sentiment analysis into SVM-based machine learning approaches and generalized autoregressive conditional heteroskedasticity modeling. Computational results show that the proposed approach has a higher classification accuracy than the semantic approach. Xia et al. [189] proposed a distantly supervised lifelong learning framework for large-scale social media sentiment analysis. Experiments on nine benchmark sentiment datasets show that by learning continuously on past tasks and combining the knowledge learned from them, the proposed lifelong ensemble framework beats the traditional way of using all training data in one task, in both classification performance and computational efficiency.

2.2.3 Deep Learning Based Methods. Traditional machine learning based TSA methods can not fully utilize the contextual information of text. To solve the problem of ignoring contextual semantics, many researchers have made in-depth research on the approach of text emotional analysis based on deep learning-based technologies, including Word Embedding, Neural Networks, Knowledge Enhanced Representation, and Transfer Learning.

Word Embedding is a technology of representing words as vectors in a multi-dimensional space, where the distance and direction between vectors reflect the similarity and relationships among the corresponding words [37]. Word2Vec [120] is one of the most popular models to learn word

[19]https://nlp.stanford.edu/sentiment/treebank.html
[20]https://github.com/JULIELab/EmoBank
[21]http://mypersonality.org/wiki/doku.php?id=download_databases
[22]http://nlp.innobic.yzu.edu.tw/resources/cvat.html
[23]https://github.com/moses-smt/mosesdecoder/
[24]https://www.ldc.upenn.edu/
[25]http://www.statmt.org/wmt14/translation-task.html
[26]https://www.statmt.org/wmt16/translation-task.html
[27]https://www.kaggle.com/akshit3050/acl-2014
[28]https://github.com/siat-nlp/MAMS-for-ABSA
[29]https://gluebenchmark.com/faq

embeddings using shallow neural network. Soon afterward, GloVe [136], **Embeddings from Language Models** (**ELMo**) [137], Emo2Vec [191], and some other embedding models were proposed, which significantly improved the TSA tasks.

Typical neural networks include **Convolutional Neural Network** (**CNN**), **Recurrent Neural Network** (**RNN**), **Long Short-Term Memory** (**LSTM**), **Graph Neural Network** (**GNN**) and so on. Many scholars have proposed different deep learning based methods for social media TSA. For example, Zhao et al. [202] proposed a novel deep learning framework for product review sentiment classification which employs prevalently available ratings as weak supervision signals. Experimental results on a dataset containing 1.1M weakly labeled review sentences and 11,754 labeled review sentences from Amazon show the efficacy of the proposed framework and its superiority over baselines. On the basis of CNN model, Ling et al. [107] proposed a novel structure of a hybrid neural network model to deal with the polysemy phenomena of words and topic confusion with Weibo. Experimental results indicate that the proposed model performs better on the precision, recall, and F1-score for Weibo TSA.

The attention mechanism extends the power of neural networks, allowing for approximates of more complex functions, i.e., focusing on specific parts of the input. By using this mechanism, the performance of NLP tasks can be effectively improved, especially in TSA. Many scholars have developed attention mechanism in their research.

For example, Vaswani et al. [173] proposed a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Subsequently, many Transformer-based deep learning models are proposed for TSA of OSNs data. For example, Yin and Zhong [195] proposed a double-view **graph Transformer on text** (**TextGT**), which shows well performance in aspect-based sentiment analysis. Zheng et al. [205] proposed to model the sequential relation between two words by integrating the information from positions, directions, and content all together in a unified self-attentive framework. Experiments in TSA tasks show that the sequential relation modeling helps self-attention networks outperform existing approaches. Bai et al. [10] investigated a novel **relational graph attention network** (**RGAT**) that integrates typed syntactic dependency information. Experiments show that these proposed methods can effectively improve sentiment classification performances.

Pre-trained language models have been widely used in TSA. Devlin and his colleagues from Google proposed a new language representation model, named **Bidirectional Encoder Representations from Transformers** (**BERT**) [42], which is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. It obtained new state-of-the-art results on eleven natural language processing tasks. Some recent works have focused on injecting sentiment knowledge into pre-trained models [102, 169, 199]. For example, Li et al. [102] proposed a novel knowledge enhanced model, which simultaneously incorporates commonsense and sentiment knowledge as external knowledge. Experiments demonstrate the usefulness of both external knowledge sources in TSA tasks.

Transfer learning aims at improving the performance of target learners on target domains by transferring the knowledge contained in different but related source domains [210]. Due to its superiority in learning knowledge from limited training data, many researchers introduced transfer learning into TSA tasks [29, 109]. For instance, Pham et al. [138] introduced UniBridge, a comprehensive approach developed to improve the effectiveness of Cross-Lingual Transfer Learning, particularly in languages with limited resources.

In recent years, **Large language models** (**LLMs**), such as **Generative Pre-trained Transformer** (**GPT**-3) [24] and GPT-4 [129], are making new waves in the field of NLP tasks and

beyond, due to their emergent ability and generalizability [203]. A comprehensive overview of LLMs in TSA can be found in [200].

2.3 Applications of Computational Sentiment Analysis

With the fast development of the Internet, most news agencies and media organizations have shifted their focuses on publishing online news and articles. Furthermore, more and more people express their sentiments and opinions towards various entities like products, movies, services, social and political events, and so on. through their online posts, reviews, blogs, and articles. The exponential proliferation of opinionated online texts has promoted many applications of computational sentiment analysis. Some of them are briefly introduced below.

2.3.1 Market Prediction. Behavior economics believe that financial decisions can be significantly impacted by people's emotions [126]. Public sentiments and opinions expressed in online news, tweets, and messages towards various markets like stock, currency, and futures can be automatically crawled and computationally analyzed to facilitate market prediction [17, 40, 44, 46, 145, 159]. For example, Bollen et al. [17] investigated the relation between public sentiment in tweets and the **Dow Jones Industrial Average** (**DJIA**) index over time series sentiment analysis, while their results suggest the accuracy of DJIA predications can be significantly improved by the inclusion of such a public emotional dimension. Siganos et al. [159] exploited the sentiment analysis of Facebook's Gross National Happiness Index[30] to investigate the correlation between daily public sentiment and stock returns, trading volume and stock price volatility across twenty international markets. A positive relation between sentiment on Facebook and stock market returns had been observed, yet sentiment is negatively related to trading volume and volatility.

2.3.2 Management Intelligence. Sentiment analysis can prompt business and governance intelligence. For companies, sentiment analysis can help to manage brand images [110], product sales [6] and customer relations [9]; While for governments, it can be used to track and predict public sentiments towards various political issues, like election campaigns [86], policy regulations [27], and social events [2]. For example, Liu et al. [110] developed a platform to automatically extract brand-related tweets by using the LDA technique and classify their sentiments by using the Stanford CoreNLP software package [113]. More than 1.7 million tweets for 20 brands across 5 industries have been analyzed. Their findings show that band-related tweets are commonly charged with emotions, yet negative tweets are much more than positive ones. Their work can help brand managers recognize consumers' overall attitudes toward their companies and products, so as to conduct public relations and targeted advertisements timely and efficiently. Paltoglou [132] examined several millions of tweets over a one-month time span to explore whether sentiment analysis for tweets can be used for "event detection"; While the answer is confirmative, the sampling size as well as the type of an event also play important roles.

2.3.3 Public Health. Analyzing online public sentiment related to their health status and medical treatment is becoming an important tool for health and medical informatics [4, 90, 128, 134, 146]. Many patients tend to be affected by serious psychological disorders, like stress, frustration, depression, during their medical treatment and would like to take participation in some online virtual communities where they can exchange messages regarding their treatment, seeking and giving encouragement. It is thus of great importance for doctors, nurses, assistants, and families to be able to automatically detect and track patients' emotions from their online posts. For example, Paul and Dredze [134] analyzed over 1.5 million health related tweets for syndromic

[30] https://apps.facebook.com/gnh_index

surveillance, like monitoring geographic behavioral risk factors, disease distribution, and medication usage. Korkontzelos et al. [90] investigated whether sentiment analysis can help to improve the identification of *adverse drug reactions* for pharmacovigilance, based on the hypothesis that such mentions in online messages could be often charged with negative emotions. Rodrigues et al. [146] developed a sentiment analysis tool, called SentiHealth-Cancer, for detecting the emotional state of patients in Brazilian online cancer communities from their Portuguese posts.

During the COVID-19 pandemic, OSNs have become key platforms for the public to gather information and express their emotions and opinions. Rapid detection, analysis, and response to public concerns, emotions, and attitudes around COVID-19, can reduce the spread of fear and uncertainty information, help governments, institutions, and individuals to cope with the pandemic and relevant public opinion, promote better communication between the government and the public with the health, and enhance the public's trust in the public health measures [3, 75, 119, 141, 163]. For example, Hung et al. [75] conducted sentiment analysis and geographical analysis of related tweets and severally reached the top five states where the public expressed the most negative and the most positive emotions. Zhao et al. [204] used Weibo to measure the public's emotions change to public health emergencies, and found that emotional tendency on hot topics related to COVID-19 changed from negative to neutral, while positive emotions increased overall. Sun et al. [163] constructed a large-scale COVID-19 dataset from Weibo and proposed a dual COnsistency-enhanced semi-superVIseD network for Sentiment Analysis (COVID-SA). Mendes et al. [119] developed a baseline system based on modern NLP methods for human-in-the-loop fact-checking in the domain of COVID-19 treatments. Angelopoulou et al. [3] applied a bigram graph-based approach to visualize the most frequent topics in X discussions during the COVID-19 pandemic, which can offer a novel and potentially valuable contribution to public policymakers.

2.3.4 Education. Sentiment analysis also plays a vital role in the education domain. Analyzing students' feedback and emotions toward teaching and learning practices can help in evaluating learning management systems (e.g., Canvas, Moodle), online teaching tools (e.g., MOOCs), pedagogical procedures, courses and enhancing learning-teaching practices [36, 41, 48, 60, 155, 208]. For example, Edalati et al. [48] analyzed students' feedback collected from a MOOC platform by employing different conventional machine learning/deep learning models. Gkontzis et al. [60] implemented classic text mining techniques to analyze the activity, polarity, and emotions of tutors and students in order to predict students' grades. Cunningham-Nelson et al. [36] used machine learning and text analysis techniques to identify the sentiment of the student comments, and the information was then presented visually. This automated methodology provides educators with a simple, systematic way to monitor their courses and make pedagogically sound decisions on teaching strategies. Moreover, some sentiment analysis systems are developed for analyzing written student course feedback (Palaute) [64] or predicting students' pass or fail outcomes (SAPP) [92].

2.3.5 Daily Life and Entertainment. Sentiment analysis could help to characterize a personal trait to some extent, which can improve personal daily Life and entertainment experience by inferring one's potential attitude to some item [14, 57, 93, 144]. For example, García-Cumbreras et al. [57] performed experiments on a subset of the **Internet Movie Database** (**IMDb**) and analyzed the relation between the movie ratings and comments for different users. Their results can help to improve the movie recommendation accuracy for the audiences. Bauman et al. [14] proposed a **Sentiment Utility Logistic Model** (**SULM**), which integrates aspect-level sentiment analysis from user reviews and recommends favorable items together with the most important aspects for different users to enhance user experience.

2.3.6 Challenges Faced in Real-World Scenarios. As in the aforementioned brief discussions, sentiment analysis, as a powerful tool for automatically extracting opinions and emotions from diverse online texts, has been becoming an important building block of various applications in many domains. However, it comes with diverse challenges due to the complex nature of the real world and its limitations. Below are the key challenges faced when applying sentiment analysis in real-world scenarios:

- ***Data Quality and Reliability:*** OSNs often contain a mix of relevant and irrelevant information. Filtering out noise to extract meaningful insights is challenging. In addition, public sentiment is particularly vulnerable to fake news and misinformation, which can skew sentiment analysis results.
- ***Multimodal Data Sources:*** Public sentiment is expressed across various platforms, including social media, forums, MOOCs, product reviews, and news articles. Integrating and analyzing these diverse data sources is challenging. Moreover, Different sources may provide conflicting sentiment signals (e.g., positive news articles but negative social media posts), making it difficult to derive a unified sentiment result for real scenario applications.
- ***Real-Time Analysis:*** The real world is always changing and evolving dynamically, for example, financial markets operate in real-time, disease epidemiological data is updated in seconds, with sentiments changing rapidly. Delays in sentiment analysis can lead to outdated insights, rendering them useless for timely decision-making. Robust computational resources and efficient algorithms are needed for analyzing large volumes of streaming data (e.g., tweets, news articles) in real-time.
- ***Short-Term vs. Long-Term Sentiment:*** Public sentiment often fluctuates in the short term due to events like financial market volatility, infectious disease outbreaks, sports results, and entertainment activities, but long-term trends are more relevant for strategic decision-making. Distinguishing between the two is a significant challenge. Sudden events (e.g., geopolitical crises, policy changes, and public health announcements) can cause abrupt sentiment shifts, making it difficult to maintain consistent analysis.
- ***Ethical and Privacy Concerns:*** Many texts often contain personal and sensitive information, such as personal financial status, medical conditions, treatments, and patient experiences. Analyzing such data raises privacy and ethical concerns. For example, public health and daily life applications must comply with strict **regulations like the Health Insurance Portability and Accountability Act** (**HIPAA**) in the U.S. or the **General Data Protection Regulation** (**GDPR**) in the EU, which govern the use of personal health data. Financial institutions in the EU must ensure that sentiment analysis applications comply with regulations like the **Markets in Financial Instruments Directive** (**MiFID**) II, and others. Moreover, sentiment analysis could inadvertently reinforce stigma or discrimination against certain groups (e.g., people with mental health conditions or infectious diseases).

Opinions and sentiments of online texts can not only be analyzed to enhance application performance but also can impact the information propagation process in OSNs and even lead to the proliferation of more opinionated online contents, which we call the *sentiment diffusion* phenomenon. While the evolution of online information is dependent on its content gene, it may also depend on network structures and dynamics. In the next section, we briefly review some basics of OSNs.

3 Online Social Network Analysis

3.1 Components and Functionalities of Online Social Networks

The website *Sixdegrees.com* launched in 1997 has been commonly regarded as the first OSN [19]. Since then, lots of OSNs have emerged and evolved, and some become much more popular and

even dominant in the market. While many services of social networks are first web-based, more and more social networks embrace the era of the mobile Internet and provide smartphone-based services [114]. Basically, OSNs can be divided into four types, namely, private, business, general, and special interest, according to their focuses and usages [69]. For example, LinkedIn focuses on job and headhunting via establishing professional contacts; Facebook is welcome by individuals for sharing their daily lives.

Given their different representations and applications, all OSNs contain four basic components, that is, *users*, *contents*, *relations*, and *actions*. A user can be a person, company, or organization that has registered, via either anonymous or verified profile, in a social network to access its services. A user can create his own contents or share them with others, like texts, slides, speeches, and videos, within his registered social network. The relations describe how users connect with each other, either directly or indirectly. For example, the *follower-followee* relation in X means that the content created by a followee can be automatically delivered to his follower. Actions can happen between a user and his created or received contents, like withdrawing a post on Facebook, or happen between a user and another user, like unsubscribing a followee in X.

Although the participants and contents could vary much in different OSNs, two main functionalities, namely, *exchanging information,* and *establishing relation*, can be identified for all social networks. A user can post his own or repost others' articles, comments or news that are readable by his subscribed followers or friends, by which different information can be shared and exchanged among users. Besides the so-called ***user generated contents*** (**UGCs**) by individuals, the online articles published by many traditional media agencies can also be reposted or shared in different social networks through their sharing interfaces. Establishing a new relationship can be done implicitly or explicitly, and can create directional or bidirectional relations among users. For example, following a Weibo user does not need his consent, which creates a directional relation; While making a new friend in WeChat generally requires mutual consent to create a bidirectional relation. On the other hand, an existing relation can also be mutually released or unidirectionally blocked.

It is the users and their relations that build the structure of an OSN. We next briefly review some research findings on the OSN structure.

3.2 Structure of Online Social Networks

The topic of research on the structure of OSNs consists of two main parts: the structural properties of OSNs and network modeling. Structural properties analysis focuses on the features of the number of nodes and edges a network contains. Network modeling studies the formation mechanism of a network.

3.2.1 Structural Properties Analysis. The most intuitive way to understand a social network is to observe its topology. In the 1930s, Moreno first introduced a graph-theoretic approach to the analysis of human social relationships. Then, a lot of work had focused on studying the statistical characteristics of social networks [12, 121, 183]. Milgram [121] verified the theory of "six degrees of separation" through an e-mail delivery experiment, which revealed that anyone in the world can be connected to any other one in just six steps. It is also referred to as the small-world phenomenon. Watts and Strogatz [183] subsequently introduced the "small-world" model of networks, which describes the clustering and short separations of nodes found in many real-life networks. Next year, Barabási and Albert [12] found that most users in the network have fewer connections, while a few users act as "highly connected hubs", namely, the scale-free phenomenon. In scale-free networks, the degree distribution follows the Yule-Simon distribution, a power law relationship defined by $P(k) \sim k^{-\gamma}$, where the probability $P(k)$ that a node's degree equal to k was roughly proportional to $k^{-\gamma}$. For most real networks, the value of γ may vary approximately from 2 to 3.

Centrality is a widely used indicator to measure the importance of users. Researchers have defined several centrality measures, like degree centrality, betweenness centrality, closeness centrality, Pagerank centrality [23], and k-core [89]. Newman [125] analyzed the characteristics and applicable occasions of various centralities. Kitsak et al. [89] found that k-core is more suitable for describing the centrality and influence of nodes than betweenness and degree centrality. Holthoefer et al. [18] chose points with different core numbers as the initial nodes in the diffusion model and found that nodes with larger core numbers have a greater role in blocking information transmission. Zhou et al. [207] found that nodes' future centrality is highly predictable due to the natural social behavior of people through real trace-driven simulations. To know how general is the game-theoretic approach to centrality and what distinguishes some game-theoretic centralities from others, Skibski et al. [160] provided the first axiomatic characterization of game-theoretic centralities.

The assortativity coefficient r of the network reflects the degree of interconnection between similar nodes in the network [124]. The range of r is $[-1, 1]$. $r > 0$ indicates that the degrees of interconnected nodes in the network are positively correlated, namely, nodes with large degrees tend to be connected to nodes with large degrees, while neighboring nodes of nodes with small degrees generally have smaller degrees. Newman [124] found that most social networks are assortative networks. Gong et al. [62] studied the data of Google+ from July 6 to October 11, 2011, and found that the assortativity coefficient has an evolutionary law from positive to negative. Bollen et al. [16] investigated the correlation between the assortativity of real social networks and OSNs. They tracked X records for 6 months and found that the assortativity changes in OSNs have a certain impact on real social networks.

Community detection is also an important issue for network structural properties analysis [85]. Generally, a community is composed of a group of nodes that are closely connected, and the nodes between the communities are loosely connected [143]. A large number of community detection algorithms have been proposed, which can be categorized as probabilistic graphical models-based methods (e.g., Stochastic Block Model-based Methods [73], Topic Model-based Methods [198], Matrix Factorization-based Methods [97], Markov Random Field-based Methods [67], Factor Graph-based Methods [194]) and deep learning-based methods (e.g., Auto-Encoder-based Methods [193], Generative Adversarial Network-based Methods [81], Graph Convolutional Network-based Methods [84]).

3.2.2 Network Modeling. There are two types of social network modeling methods, namely, traditional generative methods and deep learning-based methods. The generative model uses statistical modeling methods, which simplifies the network structure generation process into a series of basic probabilistic steps. In this case, the network is constructed by explicitly adding new nodes to a network with a specific connection mechanism, which can vividly simulate human social behavior. The ER random graph model proposed by Erdős and Rényi [131] is the first simple random graph model, which can be generated by repeatedly connecting any two node pairs with a specific probability. Its average path length is small, which is in common with the real world. However, its other properties, such as clustering coefficient and degree distribution are very different from the real world. The WS small-world model, described by Watts and Strogatz [183], is a classic and important generative model, which offers a means of moving between the regular lattice networks and the stochastic networks. A Small-world network can be formed by adding a small number of random connections to a lattice. This type of network has a high clustering coefficient and a small average path length. The BA scale-free model [12], described by Barabási and Albert, is a preferential attachment model. It can be constructed dynamically by adding new nodes to a pre-existing network and connecting to each of the original nodes with a probability proportional to the degree

the original nodes already had. Leskovec et al. [100] proposed a generative model for networks, namely, Kronecker graph, that is both mathematically tractable and can generate networks that have the same structural properties as real networks.

Recently, deep learning based methods have been intensely studied. For example, DeepGMG [105] generates graphs by using a sequential process that generates one node at a time and connects each node to the existing partial graph by creating edges one by one. The decision process is conditioned on the node states and the graph state. GraphRNN [196] can effectively generate networks by training on a representative set of graphs and decomposing the graph generation process into a sequence of node and edge formations.

4 Information Diffusion Models

Information diffusion describes the time series process in which a piece of information, like an article or articles describing a similar topic, is communicated among the users of a social network. Although the studies of information diffusion in the physical social world might be dated back to the middle of the 20th century [63, 147], they have again attracted lots of research efforts in the cyber-social world in the last decade [56, 65, 179]. Generally, the existing information diffusion models can be classified into two categories: graph models and non-graph models [65].

4.1 Graph Models

In graph models, an underlying graph structure $G = (V, E, w)$ is assumed, where a node $v \in V$ stands for a user, an edge $(u, v) \in E$, either unidirectional or bidirectional, for the relation between u and v, and $w_{(u,v)}$ for the edge weight. At first, one or more nodes, or so-called *early adopters* [147], initiate the information diffusion by publishing a piece of information that can then be automatically delivered to the publishers' subscribers or friends through OSN services, and can then further reach more users who are connected to the users who relaying such information.[31] Actions of whether or not to relay a piece of information are taken by an individual node. Normally, actions take effect immediately and are generally assumed nonreversible. The process continues, and the information diffuses; While it terminates when no new node relays the information. Common graph models include the threshold models and the cascade models.

4.1.1 Threshold Models. In threshold models, each edge between user u and v is associated with a weight $w(u, v)$, and a threshold θ_u is used to predict whether a user u would relay a piece of information. Some examples include the LT model [63] and the ***Majority Threshold*** **(MT)** model [135]. In the LT model, a user u would relay the information, if $\sum_{v \in \mathcal{N}_u} w_{(v,u)} \geq \theta_u$, where $\mathcal{N}_u$ is the neighbors of u who have relayed the information. That is, the user u would relay the information, if there are enough of its neighbors that have done so. The MT model adopts a simple voting rule: $\theta_u = \frac{1}{2}|\mathcal{N}_u|$ and $w_{(v,u)} = 1$. That is, for a user u, if the majority of its neighbors have relayed the information, in other words, it will also do it.

Considering an instance of this model, as shown in Figure 4(a)–(c), each node is assigned a threshold random and uniformly in $[0, 1]$, In step 0, two yellow dotted nodes C and D are initially activated, and other nodes keep inactive. In step 1 (Figure 4(b)), node C activates A $(0.2 \geq 0.1)$, node D also activates node F $(0.4 \geq 0.3)$. Thus there are four active nodes and they are able to activate node I $(0.3 + 0.5 \geq 0.2)$ and E $(0.4 + 0.2 \geq 0.5)$. The diffusion process terminates when no more new active node exists.

[31] Here, we borrow the term "relay" commonly used in the computer networking domain to indicate that one distributes, post, and/or share a piece of information received from others.

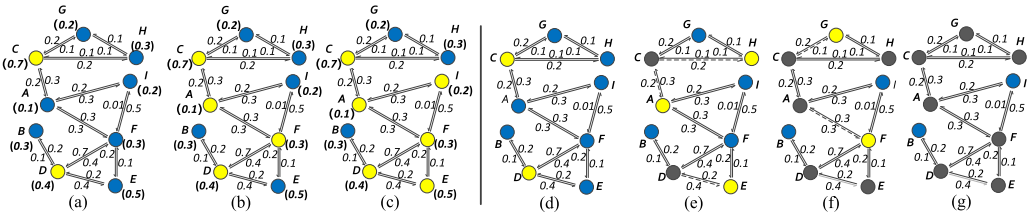


Fig. 4. Illustration of the **linear threshold (LT)** model and the **independent cascade (IC)** model [156].

4.1.2 Cascade Models. The cascade models are from the senders' viewpoints. If a node u has relayed a piece of information, it is with an activation probability p_u to successfully prompt its neighbors to also relay the information. Some examples include the IC model [61] and the **decreasing cascade** (**DC**) model [87]. In the IC model, the activation probability is a constant for each node and is also not dependent on the diffusion history. In the DC model, the activation probability is not a constant but dependent on the diffusion history. An example of the IC model is shown in Figure 4(d)–(g). In step 0, nodes C and D are activated and each of them has a chance to activate their neighbors (A, G, H) and (B, E, F), respectively. In step 1 (Figure 4(e)), only A, E, and H are successfully activated, and the initial active nodes (C and D) change to gray, since they have no chance to activate other nodes. In step 2 (Figure 4(f)), G and F are activated, and the previous active nodes (A, E, and H) change to gray. In step 3 (Figure 4(g)), G can not activate any nodes, since its neighbors C and H are already active. Meanwhile, F fails to activate I. Thus the diffusion process terminates because there is no more new node that can be activated.

4.2 Non-graph Models

Non-graph models, which originated from epidemic disease studies to model epidemiological processes [88], do not assume a graph structure to describe an information diffusion process. Instead, they specify several distinct states within which a user can stay. A user can stay within only one state at a time, however, it can transit to another state with some probability. Furthermore, each user is assumed to be connected to another with some probability, which results in random connections inside the population. The most well-known models include the SI, SIR, SIS, SIRS, and SEIR models [1, 7, 55, 70, 88].

In the SIR model, three states are: *Susceptible* (S), *Infected* (I), and *Recovered* (R). Apart from their literal meanings, they can be explained in the context of information diffusion in OSNs as follows. A susceptible user could be the one who has read but not yet relayed the information piece. An infected user could be the one who has not read but easily to receive the information piece. A recovered user could be the one who has read but decided not to relay the information piece. If a user has transited into the recovered state, it will always stay in it. In the SIS model, an infected user can transit back to the susceptible states with some probability. In the SIRS model, the recovered individuals may lose immunity and return to the susceptible state again. In the SEIR model, a new state *Exposed* (those who have been infected but are not yet infectious) is introduced.

Figure 5 illustrates the state dynamics of the SIS, SIR, SIRS, and SEIR models with transition rates. Note that the transition rates of a user can be dependent on his encountered users as well as their states.

Recently, both graph models and non-graph models for information diffusion have been increasingly studied. With the development of sentiment analysis techniques, some studies have started researching how the opinions and sentiments embedded in a piece of information would impact its diffusion and evolution in OSNs. We next review the recent advances in this field.

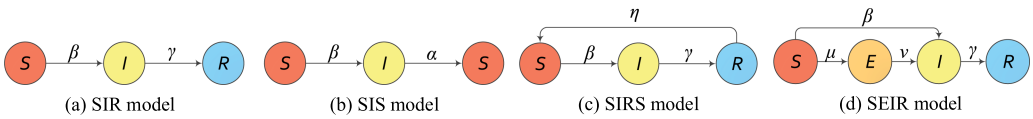


Fig. 5. Illustration of the classic non-graph models.

5 Sentiment Diffusion Models in Online Social Networks

5.1 Diffusion Models Focusing on UGC and User Information

To understand what causes and how sentiment diffusion in OSNs, a lot of studies that focused on analyzing UGCs and user information have been conducted, which are summarized in Table 3.

5.1.1 UGC-Only Based Studies. In OSNs, it was found that a negative comment is more likely to be posted after a sequence of five negative messages rather than after four such posts. In this case, some researchers considered the UGC in their sentiment diffusion models. For example, Chmiel et al. [31] studied the statistical behavior of this phenomenon and proposed a data-driven binary model of emotional online discussion threads. The sentiment ordered patterns predicted by the model have been found in the Blog06 dataset. Stieglitz et al. [162] applied regression techniques to examine whether tweet sentiment has an effect on how often a tweet has been retweeted. Based on a dataset of 64,431 political tweets, they found a positive relationship between the quantity of words indicating affective dimensions, including positive and negative emotions associated with certain political parties or politicians, in a tweet and its retweet rate. Cu et al. [35] collected 154 new digital products and studied the influence of textual UGC sentiment on new product diffusion. Their study found that the volume of posts and UGC Sentiment have a dynamic impact on the diffusion of digital products. He et al. [68] proposed a new statistical model H2 to depict the evolution of temporal network, which is governed by the competition of homophily and homeostasis. Evaluation on X data shows that H2 model can capture long-range sentiment dynamics and lower the error rate in sentiment prediction. Moreover, users in negative emotions are found more influenced by external shock events. Hatua et al. [66] described a novel model that treats information diffusion on OSNs as a multivariate time series problem. In their model, the volume of tweets, the sentiment of tweets and the influence of tweets are analyzed to characterize, predict, and quantify the impact of postings, tweets, messages and so on. on OSNs. Berry [15] examined the diffusion of information considering Ekman's six sentiment categories. He retrieved X data of four different topics, including climate change, the U.S. election, crowdfunding, and gun control, and found that there will be differences in the diffusion rates of tweets considering different sentiments and topics. Overall, the results suggested that the broad categories of positive and negative sentiment are too aggregate to really explain when information is likely to be diffused through retweets.

5.1.2 UGC and User Information Based Studies. User Information is also important for sentiment diffusion. For example, Naveed et al. [123] trained a prediction model to forecast for a given tweet its likelihood of being retweeted based on its contents. To capture the content they used low-level features such as the presence of URLs, hashtags, usernames, question and exclamation marks, emoticons, positive and negative words, as well as high-level features such as sentiments and latent topics. Based on the retweeting behavior of X users they observed, they found the tendency that bad news seem to travel fast on X. Using a unique data set of content information and user information collected from YouTube, Susarla et al. [165] found that social interactions were influential not only in determining which videos become successful but also in the magnitude of that impact. They found evidence for a number of mechanisms by which social influence is transmitted, such as (i) a preference for conformity and homophily and (ii) the role of

Table 3. Sentiment Diffusion Models in OSNs

Type	Ref.	Methods/Model	Data Source	Advantages
UGC and UI Based Studies	[31]	LMSP	BBC Forum, Digg, Blog06	The ordered patterns predicted have been found in real world dataset
	[35]	VAR model	GameSpot	The first one to claim a dynamic relationship between sentiment and new product diffusion.
	[68]	Statistical model	X	Has practical significance for those who supervise and guide health sentiments in online communities
	[66]	ARIMA, LSTM	X	Achieves far better performance and hold great potential to be utilized for real-world applications
	[15]	Text analysis	X	Can help to explain why prior diffusion studies have found conflicting positivity and negativity biases
	[123]	ANEW, LR	X	Possible to predict which tweets are retweeted
	[165]	H-T estimation	YouTube	Robust to potential self-selection
	[80]	SentiStrength	X	Can predict viral tweets with high accuracy
	[33]	LIWC	Facebook	Can determine what semantic categories are susceptible to influence between social contacts
	[209]	LDA, DL	Weibo	Can predict users' adoption behaviors among different periods in the real-world.
External User Relations Based Studies	[30]	IC-N,	NetHEPT	The first to incorporate negative opinions into the influence cascade model
	[197]	OC	NetHEPT, WikiVote	Outperforms several other heuristics both in quality and running time
	[98]	ISC	X	Improves the state-of-the-art of user modeling in sentiment diffusion
	[177]	EIC	Weibo	Achieves performance improvements compared with two baseline methods
	[190]	EIC-T	X	Shows better performance than other models of emotion contagion
	[180]	MMC	–	Helps better design effective counter-measures to inhibit the epidemic propagation in human society
	[72]	Dynamic network	BBC, Digg	Allows for the precise study of dissemination patterns and associated network effects
	[175]	LT-A	Epinions, Slashdot	Can achieve higher influence spread and is suitable for maximizing the positive influence in viral marketing
Internal State Transitions Based Studies	[71]	SISs	Framingham Heart Study	Provides a theoretical framework for studying the interpersonal spread of emotions
	[111]	SOSa-SPSa	–	Can be used to study the investor sentiment contagion in the field of behavioral finance
	[161]	Modified SOSa-SPSa	–	Can vividly depict the entire process of sentiment contagion
	[178]	ESIS	Weibo	Provides a better understanding of retweeting behaviors
	[47]	GCHMSM	MIT-SE	Provides a study of people's negative sentiment contamination
	[184]	IC,LT	Facebook	Provides a study of the propagation of both positive and negative information in OSNs

social networks in guiding opinion formation and directing product search and discovery. They also studied the problem of distinguishing between social contagion and user heterogeneity in the diffusion process. Jenders et al. [80] addressed important questions concerning the spread of information on X. By analyzing obvious features, e.g., number of followers, tweet length, number of hashtags, number of mentions, etc., and latent features, such as the sentiment and emotional divergence contained in a tweet, the most impactful features were found and a learning model that predicts viral tweets with high accuracy was proposed. Coviello et al. [33] developed a nonexperimental, noninvasive method to detect and quantify contagion of semantic expression in OSNs. Geographic aggregation and instrumental variables regression are combined to measure the effect of an exogenous variable on an individual's expression and the influence of this change on the expression of others to whom that individual is socially connected. The model can determine

what semantic categories are susceptible to influence between social contacts by estimating how an individual's usage of a semantic category is affected by her friends' usage of the same category. Zhou et al. [209] constructed the **Dynamically Socialized User Networking** (**DSUN**) model and sentiment-LDA topic model, which considers a series of social factors, including user interests and social roles, semantic topics, and sentiment polarities. They integrated the models together to facilitate the information diffusion process. A **Periodic-Aware Intelligent Prediction Model** (**PAIPM**) for information diffusion within a weighted network model is then designed to detect and classify different periods into the so-called routine period and emergency period. Finally, a deep learning scheme based on multi-factor analysis is developed for adoption behavior prediction within the identified different periods. Experiment results conducted on a real-world dataset demonstrated that the proposed model could explain the real-world information diffusion process effectively.

5.2 Diffusion Models Focusing on External User Relations

As discussed in Section 4, classic graph-based information diffusion models have focused on the external user-user relations (i.e., edges in the graph of a social network) that would impact the information diffusion, yet they have ignored the opinions and/or sentiments embedded in a piece of information that could also play an important role in its diffusion process. Some recent studies have been conducted on how to define and include sentiment-related parameters in these classic cascade and threshold diffusion models.

Chen et al. [30] proposed an ***independent cascade model with negative opinions*** (**IC-N**), which extends the IC model by adding a parameter called *quality factor* q to reflect negative opinions of users. They first defined an *influence graph* $G_{icn} = (V, E, p, q)$: Each node u has two states, namely, *active* and *inactive*,[32] and three sentiments, namely, *neutral*, *positive*, and *negative*. Furthermore, each node can stay in only one state and hold one sentiment at a time step. Each directional edge (u, v) from u to v is associated with an independent *propagation probability* p, which indicates the likelihood of an active user u to activate an inactive node v.[33] Furthermore, if v is activated by u with a negative sentiment, then v also holds a negative sentiment. Otherwise, if v is activated by u with a positive sentiment, then v may become positive with probability q, while becoming negative with probability $1 - q$. They justified the use of the quality factor q from two considerations: The diffused information, like products and services, could be inherent with some imperfectness that would be later on noticed by a recipient and change his sentiment. Thus the quality factor q can reflect the quality of the information, other than the property of the social network. Further, they regarded that negative and positive influence are asymmetric and negative one is more dominant, as a negative user u can activate a negative user v with probability one, yet a positive u can activate a positive v with only probability q.

The proposed diffusion model has made several simplifications: A newly activated user holds the same opinion of his activating neighbor, which might not be the case in real life as different neighbors might exert different influences to the user. Also, the model assumed that an opinion is discrete and countable from a limited set. So each user would hold the same opinion all the time, without any quantitative change during the diffusion process. Yet an opinion might be defined as a continuous quantity. In real life, the opinion towards a particular product or service might be different across users. Some users like it very much; While some others only slightly like it. So the

[32]We follow the term "active" and "inactive" state used in the literature, where an active (inactive) user is the one who has (has not) received and consumed the diffused information in our context.

[33]We follow the term "activate" to indicate that an active user has successfully relayed the information to an inactive user in our context.

opinion of a user could be better described by a continuous quantity. Furthermore, a continuous opinion of a user could be changed by any amount of either positive increasement or negative decreasement, when considering the differences of the influenceability and opinion intensity among his neighbors.

To address these concerns, Zhang et al. [197] proposed an ***opinion-based cascading*** (**OC**) model, and Li et al. [104] proposed a ***linear threshold with instant opinion*** (**LT-IO**) model. The two models share similar structures, and we next detail the OC model as an example. The OC model [197] extends the classic threshold models by including a new parameter, called *opinion indicator* $o_u \in [-1, 1]$ for each user u, to reflect his opinion towards the diffused content. Similar to the LT model, an inactive user u is activated, if $\sum_{v \in \mathcal{N}_u^{(a)}} w_{(v,u)} \geq \theta_u$, where $w_{(v,u)}$ is the weight of the directional edge (v, u) from v to u, and $\mathcal{N}_u^{(a)}$ the set of active neighbors of u. At the time step of u transiting from inactive to active, his opinion score is updated by $o_u = o_u + \sum_{v \in \mathcal{N}_u^{(a)}} w_{(v,u)} \times o_v$. Note that $o_u > 0$ indicates a positive opinion, $o_u = 0$ neutral, and $o_u < 0$ negative, respectively. After a user has been activated, he shall stay active and possess one of the sentiments without further changes. The authors put their information diffusion model against the virus marketing scenarios and argue that a user may have already possessed an initial opinion towards some product or service, though his opinion may be influenced and even changed according to the opinions of his active neighbors during the sentiment diffusion process.

Although using an opinion-related coefficient is able to model the sentiment diffusion process to some extent, setting its value appropriately might not be an easy task. Some researchers have argued that a single scalar coefficient might not be enough to describe the sentiment interaction among users during the information diffusion, but user types, traits, and actions could also play an important role in the diffusion process [98, 177, 190]. As argued by Lee and Lim [98], when a user u receives some content diffused from his neighbor v, his sentiment towards the content is not only dependent on the influenceability of his neighbor u but also dependent on his own intrinsic sentiment toward the diffused content. Furthermore, whether the user u is a person with a susceptible or cynical personality also impacts his sentiment polarity. Thus they proposed an ***influence-susceptibility-cynical*** (**ISC**) model to measure how a user could react to a piece of sentiment-charged information and validated their model through two real X datasets.

Wang et al. [177] proposed an *emotion-based IC model* which exploits three classes of features to determine whether a user changes his sentiment when retweeting a received tweet. The feature classes are: ***user features*** (***UF***), such as user registered profile and other demographic statistics, ***structural features*** (***SF***), such as the number of users in the path of the retweets and the number of retweeters' followers, and ***tweet features*** (***TF***), such as the tweet length and its embedded sentiments. During the sentiment diffusion of tweet messages, it is needed to decide whether a newly activated user would retweet the diffused message in their model; If so, next to determine whether he would change the sentiments of the retweeted message, as well as the sentiments of the retweets. An emotion-based IC model is then constructed based on the results of the above three actions to describe the diffusion process.

Xiong et al. [190] proposed an *emotional IC model* which includes a new network parameter, called *transmissibility*, to measure the capability of spreading emotion between two users. As illustrated in Figure 6(a), they first divide X network into four layers, namely, following layer, retweeting layer, mention layer and reply layer, each representing a user behavior in X. Two users may possess different network relations in different layers and belong to different communities. The transmissibility between two users depends on their spatio-temporal statistics, i.e., their historical behaviors and their network structures. Although the user behavior and network structure could change during the diffusion process of X messages, the transmissibility is assumed to be fixed when

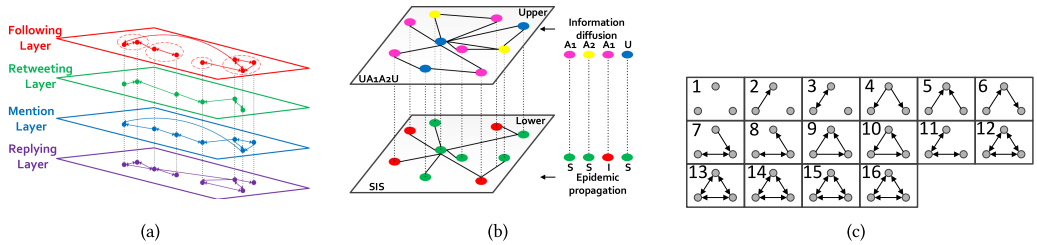


Fig. 6. (a) The multilayer structure of Twitter [190]. Layers reflect different types of user-user relations in Twitter, yet user-user relations may be different in different layers. (b) Framework of two-layered multiplex networks [180]. The upper layer represents the diffusion of positive and negative information, and the lower layer describes the epidemic propagation. (c) Isomorphic triad classes [72].

studying a particular sentiment diffusion process. An emotional IC model is then constructed based on the transmissibility matrix among users to describe the emotion diffusion process.

To explore the influence of positive and negative preventive information on epidemic propagation, Wang et al. [180] proposed a novel epidemic model based on two-layered multiplex networks, as shown in Figure 6(b). In the model, the upper layer represents a social network with positive and negative preventive information spreading competitively. Specifically, the nodes are classified into three compartments: *(1) unaware (U)*; *(2) aware of positive preventive information (A_1)*; and *(3) aware of negative preventive information (A_2)*. The lower layer denotes the physical contact network with epidemic propagation. Considering the coupling of epidemic propagation and information diffusion, there are a total of six possible states: US, UI, A_1S, A_1I, A_2S, and A_2I. Based on the **microscopic Markov chain** (**MMC**) approach, the expression of the epidemic threshold was derived, which indicates that the diffusion of positive and negative prevention information, as well as the topology of the physical contact network have a significant impact on the epidemic threshold.

Hillmann and Trier [72] used a novel dynamic network motif approach to analyze sentiment dissemination patterns in OSNs. To identify differences regarding the influence of positive and negative sentiments, sentiment-based sub-networks are used. These sub-networks are extracted by only utilizing messages with a certain sentiment type as the basis for the network topology, as illustrated in Figure 6(c). Experimental results on 5 datasets show that the emergence of digital social networks exhibits a strong tendency towards reciprocity, followed by the dominance of hierarchy as an intermediate step leading to social clustering with hubs and transitivity effects for both positive and negative sentiments to the same extent.

The aforementioned sentiment diffusion models are embedded within the unsigned social networks consisting of only positive relations, like friend or trust, among users. For an unsigned social network, the edge weight in the underlying graph $w_{(u,v)} > 0$ only indicates the influenceability of user u to user v, other than describing their relation polarity. Some social networks allow a user to explicitly tag the polarity (trust or distrust) of his relations to other users, such as in Epinions and Slashdot on the one hand. On the other hand, the relation polarity can be analyzed from his historical interactions with others [82]. Therefore, a signed OSN contains both positive (trust) and negative (distrust) relations among users.

Some researchers have studied the sentiment diffusion in signed OSNs [74, 99, 174, 175]. For example, Wang et al. [175] proposed a ***linear threshold with attitude*** (**LT-A**) sentiment diffusion model for signed OSNs. Compared with the OC graph structure, they included one more binary parameter $\rho_{(u,v)}$ to indicate the relation polarity. $\rho_{(u,v)} = 1$ indicates that user v trusts user u and would hold a same opinion of user u; $\rho_{(u,v)} = 0$ suggests that v does not trust u and would hold an opposite opinion of user u.

In [74], a ***sign-aware* cascade** (**SC**) model has been proposed for modeling the opinion diffusion in a signed OSN, which extends the classic IC model to include both trust and distrust relations among users and their impacts on positive or negative user activation. In particular, an active user can positively activate his inactive neighbors with trust relations; Otherwise, he can negatively activate his neighbors with distrust relations, that is, his opponent could be activated yet holding an opposite opinion.

Tang et al. [167] conducted a systematic investigation on the problem of sentiment diffusion in large networks. They formulated the problem as a semi-supervised learning problem and proposed a probabilistic factor graph model, named SSFG, to solve this problem. Specifically, each user's polarity and social correlation between users are modeled together as a joint probability. They assumed that the influence between the nodes only occurs within a distance of 1. Thus a user's sentiment is only influenced by himself and his followees. An efficient algorithm based on Metropolis-Hasting is presented to optimize the factor graph model. The algorithm supports both supervised learning and semi-supervised learning, thus the proposed model is able to learn a sentiment prediction model with only a small number of labeled data and a large number of unlabeled data. The proposed model was validated on a dataset from Tencent network. Experimental results show that the method can clearly improve the prediction accuracy (+3-10%) over the baseline methods.

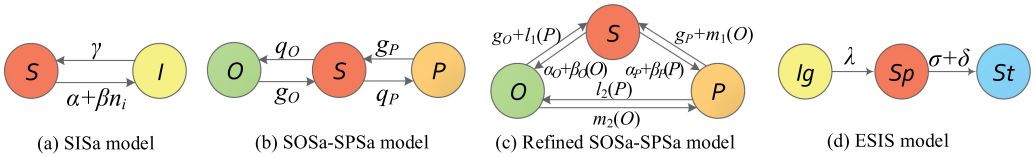
In short, the graph-based models enjoy the advantages of the detailed graph structure, which makes it possible to provide a detailed description about the full diffusion process from the uppermost originators to the bottommost receivers. The aforementioned models by augmenting the graph structure with a sentiment-related coefficient, either user-independent or user-dependent, help to predict how sentiment would spread along with the information diffusion. However, they could suffer from the veracity of the underlying graph structure in a large OSN. Also, the network dynamics and scalability may weaken their accuracy. In some cases, the diffusion details may not be the focus, but the convergence distribution, where the graph structure may not be needed but the sentiment contagion dynamics plays an important role.

5.3 Diffusion Models Focusing on Internal State Transitions

Like modeling epidemiological processes, some sentiment diffusion models focus on the internal emotion state dynamics of individual users. The internal emotional state of a user is not only dependent on the internal state transition probabilities by which a user could spontaneously change his emotional state but also related to casual user-user interactions by which opinions could be infected from one user to the other. Yet they normally do not assume a clear graph structure that directs the information transmission path, and the interactions could be random among users. To better describe sentiment-related interactions, personal traits, psychological elements, and other influential factors could be used as or included into user internal states.

Hill et al. [71] proposed a sentiment diffusion model, called SISa model, which slightly modifies the classic SIS model by allowing a user emotion to be automatically or spontaneously infected with some probability. In particular, an uninfected user can automatically become infected at a constant rate, independent of his external interactions. Though a simple extension, the SISa model actually tries to capture both internal consciousness and external transmissibility for individual sentimental dynamics. It is worth noting that the SISa model was proposed from the observations of offline social networks; While the user interaction patterns are far more complicated in OSNs. Figure 7(a) illustrates the state dynamics of the SISa model.

Liu et al. [111] regarded that the single infected state in the SIS model was not enough to describe the sentiment polarity. They divided the infected state into two opposite sentiment states, namely, optimistic and pessimistic. Based on the SISa model [71], they proposed a new SOSa-SPSa



(a) SISa model (b) SOSa-SPSa model (c) Refined SOSa-SPSa model (d) ESIS model

Fig. 7. Illustration of the SISa model [71], SOSa-SPSa model [111], refined SOSa-SPSa model [161], and ESIS model [178].

model, which includes two diffusion processes, susceptible-optimistic-susceptible and sentiment-pessimistic-susceptible, each with an automatic infection. Similar to the SISa model, the susceptible can transit into either the optimistic or the pessimistic at a certain probability. In the SOSa-SPSa model, the optimistic and the pessimistic can only transit into the susceptible but cannot transit into each other directly. Figure 7(b) illustrates the state dynamics of the SOSa-SPSa model. Song et al. [161] extended the SOSa-SPSa model by allowing the interaction between optimists and pessimists, yet the transition probabilities are different in the two transit directions, as shown in Figure 7(c).

Wang et al. [178] proposed an **emotion-based spreader-ignorant-stifler** (**ESIS**) model to predict the tweet diffusion in X. Similar to the classic SIS model, three user types are identified: A spreader posts or reposts a tweet; An ignorant does not receive the tweet; And a stifler has already received the tweet but will not repost it again. Different transition probabilities are defined: The spreading probability describes the likelihood that an ignorant becomes a spreader when encountering another spreader; The active and passive stifling probability determines whether a spreader would transit to a stifler spontaneously or when meeting another spreader. A tweet can trigger a user with different emotions (Ekman's Six Basic Emotions plus "no emotion") that could impact the spreading probability during the tweet diffusion process. Figure 7(d) illustrates the state dynamics of the ESIS model.

Considering the difference between sentiment contagion and epidemic contagion, Zhao et al. [201] defined the transition probabilities of the binary emotional state (optimism, pessimism) and established a sentiment contagion model. Transition equations are given in a homogeneous network. The equations were transformed into a differential equation about the proportion of the number of the pessimists and the optimists, and the stability of the zero solution is discussed. Numerical simulation results conducted by the Monte Carlo method show that the overall tendency of sentiment variation in the BA scale-free network is similar in the homogeneous network.

Du et al. [47] designed a ***graph-coupled hidden Markov Sentiment Model*** (**GCHMSM**) for modeling negative emotion spreading in mobile social networks, which is an extension of the GCHMM [43] designed for modeling the spread of infectious diseases by likening the sentiment as a disease. They define a *frustrating value* to measure a user's negative emotion contamination ability, which is related to the ratio of his encountered negative users. The internal emotional states of a user include both hidden and observable susceptible normal state and infected negative state. A hidden Markov model is used to model the generative process between the hidden and observable emotion states. However, the hidden emotion transition is dependent not only on his own dynamic infectious capabilities but also on his neighbors' hidden emotional states.

Wen et al. [184] extended the basic three states in the SIR model with more child states to capture diverse user behaviours in reacting to opinionated information with different polarity. They divide the recovered state into three child states, namely, *insider*, *active*, and *immunized*; and also the infected state into three child states, namely, *misled*, *contagious* and *dormant*. The external actions include whether to read the information and if so, whether to spread the information.

Martins [116] considered a more general case where the internal state of a user is no longer countable discrete. Instead, each user holds continuous opinions that could be updated via interactions with his encountered neighbors. Yet the external actions are considered as discrete actions, like approval or disapproval, which could be the probabilistic output of his internal continuous opinion distribution. Hence Martins named the model ***continuous opinions and discrete actions*** (**CODA**).

Si et al. [157] extended the CODA model to the sentiment-charged information diffusion by including two personality-related thresholds, namely, infected threshold and refractory threshold, to describe how a user would react to a piece of opinionated information; While infected threshold reflects the enthusiasm and refractory threshold the sluggish degree.

Some researchers have also taken into consideration internal user recognition factors and external mass media influences in modeling diffusion processes. For example, Lakkaraju [94] proposed a ***multi-agent multi-attitude*** (**MAMA**) model where an internal cognitive network is embedded within each user to represent cognitions and the interactions between them. Cognition refers to the user's perception toward some entity, like a topic or event diffused in the OSN; While the user's attitude towards cognition is a continuous value ranging in $[-1, +1]$. External media is introduced to the diffusion process, which functions as agenda mediators for setting topic distribution over cognitions. Yet their diffusion dynamics are based on an epidemic process via random user interactions.

6 Future Research Directions

Sentiment diffusion in OSNs is a new and emerging field, offering many directions to pursue. Researchers can continually discover and provide better solutions in this field.

To improve the present algorithms and methods, a possible way is to create synergy among the different solutions that were presented in Sections 2 to 6. Additionally, state-of-the-art sentiment classification and network structure detection algorithms could be used to develop solutions more accurately. For the empirical studies, a worthwhile direction is trying many more datasets to acquire more evidence. With more real-trace records, more interesting and valuable results may be found. Besides, some researchers have found that sentiment diffusion can affect social connection in turn, which means that social networks can be dynamically changing. Thus, the impact of such changes should be considered as a further research direction. Additional possible future research directions include modeling the process of sentiment diffusion more closely to real-life situations.

7 Conclusion

As emerging research in OSNs, sentiment diffusion has a broad application prospect and has attracted widespread attention from researchers. In this survey article, we present a survey of sentiment diffusion in OSNs from the computational perspective. First, we present the definition and classification of sentiment, introduce the most advanced computational sentiment analysis techniques, list their applications, and disclose available sentiment analysis resources. Then the main OSN components, functionalities, and structural features applied in modeling social networks are analyzed and summarized. It can be seen that these network features play an important role in OSNs, not only in analyzing the network structures, generating a more realistic network, but also in dealing with the diffusion of information. Next, a thorough overview of the information diffusion technologies in OSNs, including graph and non-graph models is made. At last, we provide a systematic review of the current state, issues, and future directions of research on sentiment diffusion in OSNs. We hope that this survey article will give relevant researchers a better understanding and new insights for further research on sentiment diffusion in OSNs.

References

[1] M. E. Alexander and S. M. Moghadas. 2005. Bifurcation analysis of an SIRS epidemic model with generalized incidence. *SIAM Journal on Applied Mathematics* 65, 5 (2005), 1794–1816. DOI: https://doi.org/10.1137/040604947

[2] Nasser Alsaedi, Pete Burnap, and Omer Rana. 2017. Can we predict a riot? Disruptive event detection using Twitter. *ACM Transactions on Internet Technology* 17, 2 (2017), 1–26. DOI: https://doi.org/10.1145/2996183

[3] Anastasia Angelopoulou, Konstantinos Mykoniatis, and Alice E. Smith. 2024. Analysis of public sentiment on COVID-19 mitigation measures in social media in the United States using machine learning. *IEEE Transactions on Computational Social Systems* 11, 1 (2024), 307–318. DOI: https://doi.org/10.1109/TCSS.2022.3214527

[4] V. S. Anoop and S. Sreelakshmi. 2023. Public discourse and sentiment during Mpox outbreak: An analysis using natural language processing. *Public Health* 218 (2023), 114–120. DOI: https://doi.org/10.1016/j.puhe.2023.02.018

[5] Oscar Araque, Ganggao Zhu, and Carlos A. Iglesias. 2019. A semantic similarity-based perspective of affect lexicons for sentiment analysis. *Knowledge-Based Systems* 165 (2019), 346–359. DOI: https://doi.org/10.1016/j.knosys.2018.12.005

[6] Nikolay Archak, Anindya Ghose, and Panagiotis G. Ipeirotis. 2011. Deriving the pricing power of product features by mining consumer reviews. *Management Science* 57, 8 (2011), 1485–1509. DOI: https://doi.org/10.1287/mnsc.1110.1370

[7] Joan L. Aron and Ira B. Schwartz. 1984. Seasonality and period-doubling bifurcations in an epidemic model. *Journal of Theoretical Biology* 110, 4 (1984), 665–679. DOI: https://doi.org/10.1016/S0022-5193(84)80150-2

[8] James R. Averill. 1980. Chapter 12 - A constructivist view of emotion. In *Proceedings of the Theories of Emotion*, Robert Plutchik and Henry Kellerman (Eds.). Academic Press, 305–339. DOI: https://doi.org/10.1016/B978-0-12-558701-3.50018-1

[9] Ayoub Bagheri, Mohamad Saraee, and Franciska De Jong. 2013. Care more about customers: Unsupervised domain-independent aspect detection for sentiment analysis of customer reviews. *Knowledge-Based Systems (Elsevier)* 52 (2013), 201–213. DOI: https://doi.org/10.1016/j.knosys.2013.08.011

[10] Xuefeng Bai, Pengbo Liu, and Yue Zhang. 2021. Investigating typed syntactic dependencies for targeted sentiment classification using graph attention neural network. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 29 (2021), 503–514. DOI: https://doi.org/10.1109/TASLP.2020.3042009

[11] Alexandra Balahur, Jesus M. Hermida, and Andres Montoyo. 2012. Building and exploiting EmotiNet, a knowledge base for emotion detection based on the appraisal theory model. *IEEE Transactions on Affective Computing* 3, 1 (Jan. 2012), 88–101. DOI: https://doi.org/10.1109/T-AFFC.2011.33

[12] Albert-László Barabási and Réka Albert. 1999. Emergence of scaling in random networks. *Science* 286, 5439 (Oct. 1999), 509–512. DOI: https://doi.org/10.1126/science.286.5439.509

[13] Lisa Feldman Barrett and James A. Russell. 1998. Independence and bipolarity in the structure of current affect. *Journal of Personality and Social Psychology* 74, 4 (1998), 967–984. DOI: https://doi.org/10.1037/0022-3514.74.4.967

[14] Konstantin Bauman, Bing Liu, and Alexander Tuzhilin. 2017. Aspect based recommendations: Recommending items with the most valuable aspects based on user reviews. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD'17)*. 717–725. DOI: https://doi.org/10.1145/3097983.3098170

[15] Colin Berry. 2020. The diffusion of information: The impact of sentiment and topic on retweets. In *Proceedings of the 2020 IEEE International Conference on Big Data (Big Data'20)*. 5636–5638. DOI: https://doi.org/10.1109/BigData50022.2020.9377772

[16] Johan Bollen, Bruno Goncalves, Guangchen Ruan, and Huina Mao. 2011. Happiness is assortative in online social networks. *Artificial Life* 17, 3 (2011), 237–251. DOI: https://doi.org/10.1162/artl_a_00034

[17] Johan Bollen, Huina Mao, and Xiaojun Zeng. 2011. Twitter mood predicts the stock market. *Journal of Computational Science (Elsevier)* 2, 1 (2011), 1–8. DOI: https://doi.org/10.1016/j.jocs.2010.12.007

[18] Javier Borge-Holthoefer and Yamir Moreno. 2012. Absence of influential spreaders in rumor dynamics. *Physical Review E* 85, 2 (Feb. 2012), 026116. DOI: https://doi.org/10.1103/PhysRevE.85.026116

[19] Danah M. Boyd and Nicole B. Ellison. 2007. Social network sites: Definition, history, and scholarship. *Journal of Computer-Mediated Communication (Wiley)* 13, 1 (2007), 210–230. DOI: https://doi.org/10.1111/j.1083-6101.2007.00393.x

[20] Ryan L. Boyd, Ashwini Ashokkumar, Sarah Seraj, and James W. Pennebaker. 2022. The development and psychometric properties of LIWC-22. *Austin, TX: University of Texas at Austin* (2022). https://www.liwc.app

[21] Margaret M. Bradley, Mark K. Greenwald, Margaret C. Petry, and Peter J. Lang. 1992. Remembering pictures: Pleasure and arousal in memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 18, 2 (1992), 379–390.

[22] Margaret M. Bradley and Peter J. Lang. 1999. *Affective Norms for English Words (ANEW): Instruction Manual and Affective Ratings*. Technical Report. Technical report C-1, the center for research in psychophysiology ....

[23] Sergey Brin and Lawrence Page. 1998. The anatomy of a large-scale hypertextual Web search engine. *Computer Networks and ISDN Systems* 30, 1 (April 1998), 107–117. DOI: https://doi.org/10.1016/S0169-7552(98)00110-X

[24] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. In *Proceedings of the 34th International Conference on Neural Information Processing Systems* (Vancouver, BC, Canada) *(NIPS'20)*. Curran Associates Inc., Red Hook, NY, USA, Article 159, 25 pages.

[25] Erik Cambria, Yang Li, Frank Z. Xing, Soujanya Poria, and Kenneth Kwok. 2020. SenticNet 6: Ensemble application of symbolic and subsymbolic AI for sentiment analysis. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management* (Virtual Event, Ireland) *(CIKM'20)*. ACM, New York, NY, USA, 105–114.

[26] E. Cambria, A. Livingston, and A. Hussian. 2012. *The Hourglass of Emotion*. Springer-Verlag. 144–157 pages.

[27] Andrea Ceron, Luigi Curini, and Stefano M. Iacus. 2016. First- and second-level agenda setting in the twittersphere: An application to the Italian political debate. *Journal of Information Technology and Politics* 13, 2 (2016), 159–174.

[28] Koyel Chakraborty, Siddhartha Bhattacharyya, and Rajib Bag. 2020. A survey of sentiment analysis from social media data. *IEEE Transactions on Computational Social Systems* 7, 2 (2020), 450–464.

[29] Jireh Yi-Le Chan, Khean Thye Bea, Steven Mun Hong Leow, Seuk Wai Phoong, and Wai Khuen Cheng. 2023. State of the art: A review of sentiment analysis based on sequential transfer learning. *Artificial Intelligence Review* 56, 1 (2023), 749–780.

[30] Wei Chen, Alex Collins, Rachel Cummings, Te Ke, Zhenming Liu, David Rincon, Xiaorui Sun, Wei Wei, Yajun Wang, and Yifei Yuan. 2011. Influence maximization in social networks when negative opinions may emerge and propagate. In *Proceedings of the 2011 SIAM International Conference on Data Mining (SDM'11)*. 379–390.

[31] Anna Chmiel and Janusz A. Hołyst. 2013. Transition due to preferential cluster growth of collective emotions in online communities. *Physical Review E* 87, 2 (2013), 1–9. DOI: https://doi.org/10.1103/PhysRevE.87.022808

[32] Marco Conti and Andrea Passarella. 2017. Online social networks and media. *Online Social Networks and Media* 1 (Jun 2017), iii–vi. DOI: https://doi.org/10.1016/S2468-6964(17)30045-9

[33] Lorenzo Coviello, James H. Fowler, and Massimo Franceschetti. 2014. Words on the web: Noninvasive detection of emotional contagion in online social networks. *Proceedings of the IEEE* 102, 12 (2014), 1911–1921.

[34] Alan S. Cowen and Dacher Keltner. 2017. Self-report captures 27 distinct categories of emotion bridged by continuous gradients. *Proceedings of the National Academy of Sciences* 114, 38 (2017), E7900–E7909.

[35] Tung Cu, Helmut Schneider, and James Van Scotter. 2016. New product diffusion: The role of sentiment content. In *Proceedings of the 2016 ACM SIGMIS Conference on Computers and People Research* (Alexandria, Virginia, USA) *(SIGMIS-CPR'16)*. Association for Computing Machinery, New York, NY, USA, 149–155. DOI: https://doi.org/10.1145/2890602.2890627

[36] Samuel Cunningham-Nelson, Mahsa Baktashmotlagh, and Wageeh Boles. 2019. Visualizing student opinion through text analysis. *IEEE Transactions on Education* 62, 4 (2019), 305–311. DOI: https://doi.org/10.1109/TE.2019.2924385

[37] Liliane Soares da Costa, Italo L. Oliveira, and Renato Fileto. 2023. Text classification using embeddings: A survey. *Knowledge and Information Systems* 65, 7 (2023), 2761–2803.

[38] Charles Darwin. 1872. *The Expression of the Emotions in Man and Animals*. Oxford University Press.

[39] Jiawen Deng and Fuji Ren. 2023. A survey of textual emotion recognition and its challenges. *IEEE Transactions on Affective Computing* 14, 1 (2023), 49–67. DOI: https://doi.org/10.1109/TAFFC.2021.3053275

[40] Xiang Deng, Vasilisa Bashlovkina, Feng Han, Simon Baumgartner, and Michael Bendersky. 2023. What do LLMs know about financial markets? A case study on reddit market sentiment analysis. In *Proceedings of the ACM Web Conference* (Austin, TX, USA) *(WWW'23 Companion)*. ACM, 107–110. DOI: https://doi.org/10.1145/3543873.3587324

[41] Charalampos Dervenis, Giannis Kanakis, and Panos Fitsilis. 2024. Sentiment analysis of student feedback: A comparative study employing lexicon and machine learning techniques. *Studies in Educational Evaluation* 83 (2024), 101406.

[42] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Association for Computational Linguistics, Minneapolis, Minnesota, 4171–4186. DOI: https://doi.org/10.18653/v1/N19-1423

[43] Wen Dong, Alex Pentland, and Katherine A. Heller. 2012. Graph-coupled HMMs for modeling the spread of infection. In *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence* (Oct 2012), 227–236. arXiv:1210.4864

[44] Ksenija Doroslovacki, Nikola Gradojevic, and Albane Christine Tarnaud. 2024. A novel market sentiment analysis model for forecasting stock and cryptocurrency returns. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 54, 9 (2024), 5248–5259. DOI: https://doi.org/10.1109/TSMC.2024.3402160

[45] Jie Du, Chi-Man Vong, and C. L. Philip Chen. 2021. Novel efficient RNN and LSTM-like architectures: Recurrent and gated broad learning systems and their applications for text classification. *IEEE Transactions on Cybernetics* 51, 3 (March 2021), 1586–1597. DOI: https://doi.org/10.1109/TCYB.2020.2969705

[46] Kelvin Du, Frank Xing, Rui Mao, and Erik Cambria. 2024. Financial sentiment analysis: Techniques and applications. *ACM Comput. Surv.* 56, 9, Article 220 (Apr 2024), 42 pages. DOI: https://doi.org/10.1145/3649451

[47] Zhanwei Du, Yongjian Yang, Qing Cai, Chijun Zhang, and Yuan Bai. 2018. Modeling and inferring mobile phone users' negative emotion spreading in social networks. *Future Generation Computer Systems (Elsevier)* 78 (2018), 933–942.
[48] Maryam Edalati, Ali Shariq Imran, Zenun Kastrati, and Sher Muhammad Daudpota. 2022. The potential of machine learning algorithms for sentiment classification of students' feedback on MOOC. In *Proceedings of the Intelligent Systems and Applications*, Kohei Arai (Ed.). Springer International Publishing, Cham, 11–22.
[49] Paul Ekman. 1992. Are there basic emotions? *Psychological Review* 99, 3 (1992), 550–553.
[50] Paul Ekman. 1999. *Basic Emotions*. John Wiley & Sons, Ltd, Chapter 3, 45–60. DOI: https://doi.org/10.1002/0470013494.ch3
[51] Andrea Esuli and Fabrizio Sebastiani. 2006. SENTIWORDNET: A publicly available lexical resource for opinion mining. In *Proceedings of the 5th International Conference on Language Resources and Evaluation (LREC'06)*. European Language Resources Association, Genoa, Italy. Retrieved from http://www.lrec-conf.org/proceedings/lrec2006/pdf/384_pdf.pdf
[52] Michael Fire, Roy Goldschmidt, and Yuval Elovici. 2014. Online social networks: Threats and solutions. *IEEE Communications Surveys & Tutorials* 16, 4 (2014), 2019–2036. DOI: https://doi.org/10.1109/COMST.2014.2321628
[53] J. Fontaine, K. Scherer, E. Roesch, and P. Ellsworth. 2007. The world of emotions is not two-dimensional. *Psychological Science* 18, 12 (2007), 1050–1057.
[54] Nico H. Frijda. 1986. *The Emotions*. Cambridge University Press.
[55] Sebastian Funk, Marcel Salathé, and Vincent A. A. Jansen. 2010. Modelling the influence of human behaviour on the spread of infectious diseases: A review. *Journal of The Royal Society Interface* 7, 50 (Sep 2010), 1247–1256.
[56] Sheng Gao, Huacan Pang, Patrick Gallinari, Jun Guo, and Nei Kato. 2017. A novel embedding method for information diffusion prediction in social network big data. *IEEE Transactions on Industrial Informatics* 13, 4 (2017), 2097–2105.
[57] Miguel Á García-Cumbreras, Arturo Montejo-Ráez, and Manuel C. Díaz-Galiano. 2013. Pessimists and optimists: Improving collaborative filtering through sentiment analysis. *Expert Systems with Applications (Elsevier)* 40, 17 (2013), 6758–6765. DOI: https://doi.org/10.1016/j.eswa.2013.06.049
[58] Lorenzo Gatti, Marco Guerini, and Marco Turchi. 2016. SentiWords: Deriving a high precision and high coverage lexicon for sentiment analysis. *IEEE Transactions on Affective Computing* 7, 4 (Oct. 2016), 409–421.
[59] Anastasia Giachanou and Fabio Crestani. 2016. Like it or not: A survey of Twitter sentiment analysis methods. *Comput. Surveys* 49, 2 (2016), 1–41. DOI: https://doi.org/10.1145/2938640
[60] Andreas F. Gkontzis, Sotiris B. Kotsiantis, Dimitris Kalles, Chris T. Panagiotakopoulos, and Vassilios S. Verykios. 2020. Polarity, emotions and online activity of students and tutors as features in predicting grades. *Intelligent Decision Technologies* 14, 3 (2020), 409–436. https://api.semanticscholar.org/CorpusID:222210435
[61] Jacob Goldenberg, Barak Libai, and Eitan Muller. 2001. Talk of the network: A complex look at the underlying process of word-of-mouth. *Marketing Letters* 12, 3 (May 2001), 211–223.
[62] Neil Zhenqiang Gong, Wenchang Xu, Ling Huang, Prateek Mittal, Emil Stefanov, Vyas Sekar, and Dawn Song. 2012. Evolution of social-attribute networks: Measurements, modeling, and implications using google+. In *Proceedings of the 2012 Internet Measurement Conference*. ACM, New York, NY, USA, 131–144. DOI: https://doi.org/10.1145/2398776.2398792
[63] Mark Granovetter. 1978. Threshold models of collective behavior. *American Journal on Sociology* 83, 6 (May 1978), 1420–1443.
[64] Niku Groöberg, Antti Knutas, Timo Hynninen, and Maija Hujala. 2021. Palaute: An online text mining tool for analyzing written student course feedback. *IEEE Access* 9 (2021), 134518–134529.
[65] Adrien Guille, Hakim Hacid, Cecile Favre, and Djamel A. Zighed. 2013. Information diffusion in online social networks: A survey. *SIGMOD Record* 42, 2 (Jul 2013), 17–28. DOI: https://doi.org/10.1145/2503792.2503797
[66] Amartya Hatua, Trung T. Nguyen, and Andrew H. Sung. 2017. Information diffusion on Twitter: Pattern recognition and prediction of volume, sentiment, and influence. In *Proceedings of the 4th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies* (Austin, Texas, USA) *(BDCAT'17)*. ACM, New York, NY, USA, 157–167.
[67] Dongxiao He, Xinxin You, Zhiyong Feng, Di Jin, Xue Yang, and Weixiong Zhang. 2018. A network-specific markov random field approach to community detection. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence, (AAAI-18), New Orleans, Louisiana, USA, February 2–7, 2018*. 306–313.
[68] Saike He, Xiaolong Zheng, and Daniel Zeng. 2017. The dynamics of health sentiments with competitive interactions in social media. In *Proceedings of the 2017 IEEE International Conference on Intelligence and Security Informatics (ISI'17)*. 101–106.
[69] Julia Heidemann, Mathias Klier, and Florian Probst. 2012. Online social networks: A survey of a global phenomenon. *Computer Networks (Elsevier)* 56, 18 (2012), 3866–3878. DOI: https://doi.org/10.1016/j.comnet.2012.08.009
[70] Herbert W. Hethcote. 2000. The mathematics of infectious diseases. *SIAM Review* 42, 4 (Jan 2000), 599–653.

[71] A. L. Hill, D. G. Rand, M. A. Nowak, and N. A. Christakis. 2010. Emotions as infectious diseases in a large social network: The sisa model. *Proceedings of the Royal Society B: Biological Sciences* 277, 1701 (2010), 3827–3835.

[72] Robert Hillmann and Matthias Trier. 2012. Dissemination patterns and associated network effects of sentiments in social networks. In *Proceedings of the 2012 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*. 511–516. DOI: https://doi.org/10.1109/ASONAM.2012.88

[73] Paul W. Holland, Kathryn Blackmond Laskey, and Samuel Leinhardt. 1983. Stochastic blockmodels: First steps. *Social Networks* 5, 2 (1983), 109–137. DOI: https://doi.org/10.1016/0378-8733(83)90021-7

[74] Maryam Hosseini-pozveh, Kamran Zamanifar, and Ahmad Reza Naghsh-nilchi. 2016. Maximizing the spread of positive influence in signed social networks. *Intelligent Data Analysis (IOS Press)* 20, 1 (2016), 199–218.

[75] Man Hung, Evelyn Lauren, Eric S Hon, Wendy C Birmingham, Julie Xu, Sharon Su, Shirley D Hon, Jungweon Park, Peter Dang, and Martin S Lipsky. 2020. Social network analysis of COVID-19 sentiments: Application of artificial intelligence. *Journal of Medical Internet Research* 22, 8 (2020), 1–13. DOI: https://doi.org/10.2196/22590

[76] Carroll E Izard. 1971. *The Face of Emotion*. Appleton-Century Crofts.

[77] Carroll E. Izard. 1977. *Human Emotions*. New York:Plenum Press.

[78] R. E. Jack, O. G. Garrod, and P. G. Schyns. 2014. Dynamic facial expressions of emotion transmit an evolving hierarchy of signals over time. *Current Biology* 24, 2 (2014), 187–192.

[79] William James. 1890. *The Principles of Psychology*. Henry Holt & Co.

[80] Maximilian Jenders, Gjergji Kasneci, and Felix Naumann. 2013. Analyzing and predicting viral tweets. In *Proceedings of the 22nd International Conference on World Wide Web (WWW'13)*. ACM. 657–664. DOI: https://doi.org/10.1145/2487788.2488017

[81] Yuting Jia, Qinqin Zhang, Weinan Zhang, and Xinbing Wang. 2019. CommunityGAN: Community detection with generative adversarial nets. In *Proceedings of the The World Wide Web Conference* (San Francisco, CA, USA) *(WWW'19)*. Association for Computing Machinery, New York, NY, USA, 784–794. DOI: https://doi.org/10.1145/3308558.3313564

[82] Wenjun Jiang, Guojun Wang, Md Zakirul Alam Bhuiyan, and Jie Wu. 2016. Understanding graph-based trust evaluation in online social networks. *Comput. Surveys* 49, 1 (May 2016), 1–35. DOI: https://doi.org/10.1145/2906151

[83] Yichuan Jiang and J. C. Jiang. 2015. Diffusion in social networks: A multiagent perspective. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 45, 2 (2015), 198–213. DOI: https://doi.org/10.1109/TSMC.2014.2339198

[84] Di Jin, Bingyi Li, Pengfei Jiao, Dongxiao He, and Hongyu Shan. 2019. Community detection via joint graph convolutional network embedding in attribute network. In *Proceedings of the Artificial Neural Networks and Machine Learning - ICANN (LNCS, Vol. 11731)*. Springer, 594–606. DOI: https://doi.org/10.1007/978-3-030-30493-5_55

[85] Di Jin, Zhizhi Yu, Pengfei Jiao, Shirui Pan, Dongxiao He, Jia Wu, Philip S. Yu, and Weixiong Zhang. 2023. A survey of community detection approaches: From statistical modeling to deep learning. *IEEE Transactions on Knowledge and Data Engineering* 35, 2 (2023), 1149–1170. DOI: https://doi.org/10.1109/TKDE.2021.3104155

[86] Andreas Jungherr. 2016. Twitter use in election campaigns: A systematic literature review. *Journal of Information Technology and Politics (Routledge)* 13, 1 (2016), 72–91. DOI: https://doi.org/10.1080/19331681.2015.1132401

[87] David Kempe, Jon Kleinberg, and Éva Tardos. 2005. Influential nodes in a diffusion model for social networks. In *Proceedings of the International Conference on Automata, Languages and Programming (ICALP), Springer LNCS*, Vol. 3580. 1127–1138.

[88] W. O. Kermack and A. G. McKendrick. 1927. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 115, 772 (Aug 1927), 700–721.

[89] Maksim Kitsak, Lazaros K. Gallos, Shlomo Havlin, Fredrik Liljeros, Lev Muchnik, H. Eugene Stanley, and Hernan A. Makse. 2010. Identification of influential spreaders in complex networks. *Nature Physics* 6, 11 (Nov. 2010), 888–893.

[90] Ioannis Korkontzelos, Azadeh Nikfarjam, Matthew Shardlow, Abeed Sarker, Sophia Ananiadou, and Graciela H. Gonzalez. 2016. Analysis of the effect of sentiment analysis on extracting adverse drug reactions from tweets and forum posts. *Journal of Biomedical Informatics (Elsevier)* 62 (2016), 148–158. DOI: https://doi.org/10.1016/j.jbi.2016.06.007

[91] S. D. Kreibig, F. H. Wilhelm, and W. T. Roth. 2007. Cardio-vascular, electrodermal, and respiratory response patterns to fear and sadness inducing films. *Psychophysiology* 44, 5 (2007), 787–806.

[92] Ashima Kukkar, Rajni Mohana, Aman Sharma, and Anand Nayyar. 2023. Prediction of student academic performance based on their emotional wellbeing and interaction on various e-learning platforms. *Education and Information Technologies* 28, 8 (Jan. 2023), 9655–9684. DOI: https://doi.org/10.1007/s10639-022-11573-9

[93] Sudhanshu Kumar, Kanjar De, and Partha Pratim Roy. 2020. Movie recommendation system using sentiment analysis from microblogging data. *IEEE Transactions on Computational Social Systems* 7, 4 (2020), 915–923.

[94] Kiran Lakkaraju. 2016. Modeling attitude diffusion and agenda setting: The MAMA model. *Social Network Analysis and Mining* 6, 1 (Dec 2016), 21. DOI: https://doi.org/10.1007/s13278-016-0322-4

[95] Richard S. Lazarus. 1991. Progress on a cognitive-motivational-relational theory of emotion. *The American Psychologist* 46 8 (1991), 819–34.

[96] Richard S. Lazarus and Bernice N. Lazarus. 1994. *Passion and Reason: Making Sense of Our Emotions*. Oxford University Press. DOI: https://doi.org/10.1093/oso/9780195087574.001.0001
[97] Daniel D. Lee and H. Sebastian Seung. 2000. Algorithms for non-negative matrix factorization. In *Proceedings of the 13th International Conference on Neural Information Processing Systems* (Denver, CO) *(NIPS'00)*. MIT Press, 535–541.
[98] Roy Ka-Wei Lee and Ee-Peng Lim. 2015. Measuring user influence, susceptibility and cynicalness in sentiment diffusion. In *Proceedings of the The 37th European Conference on IR Research (ECIR), Springer LNCS 9022*. 411–422.
[99] Weihua Lei, Qun Yang, and Huanhuan Wang. 2016. Positive influence maximization algorithm based on three degrees of influence. In *Proceedings of the International Conference on Intelligent Data Engineering and Automated Learning (IDEAL, also in Springer LNCS 9937)*, Vol. 2. 503–514. DOI: https://doi.org/10.1007/978-3-319-46257-8_54
[100] Jure Leskovec, Deepayan Chakrabarti, Jon Kleinberg, Christos Faloutsos, and Zoubin Ghahramani. 2010. Kronecker graphs: An approach to modeling networks. *Journal of Machine Learning Research* 33, 11 (Feb. 2010), 985–1042.
[101] Huacheng Li, Chunhe Xia, Tianbo Wang, Sheng Wen, Chao Chen, and Yang Xiang. 2021. Capturing dynamics of information diffusion in SNS: A survey of methodology and techniques. *ACM Computing Surveys* 55, 1, Article 22 (Nov 2021), 51 pages. DOI: https://doi.org/10.1145/3485273
[102] Jie Li, Xuan Li, Linmei Hu, Yirui Zhang, and Jinrui Wang. 2023. Knowledge graph enhanced language models for sentiment analysis. In *Proceedings of the 22nd International Semantic Web Conference, Athens, Greece, November 6–10, 2023, Proceedings, Part I* (Athens, Greece). Springer-Verlag, Berlin, Heidelberg, 447–464. DOI: https://doi.org/10.1007/978-3-031-47240-4_24
[103] Jun Li, Yanghui Rao, Fengmei Jin, Huijun Chen, and Xiyun Xiang. 2016. Multi-label maximum entropy model for social emotion classification over short text. *Neurocomputing* 210 (Oct. 2016), 247–256.
[104] Songsong Li, Yuqing Zhu, Deying Li, Donghyun Kim, Huan Ma, and Hejiao Huang. 2014. Influence maximization in social networks with user attitude modification. In *Proceedings of the IEEE International Conference on Communications*. 3913–3918.
[105] Yujia Li, Oriol Vinyals, Chris Dyer, Razvan Pascanu, and Peter Battaglia. 2018. Learning Deep Generative Models of Graphs. arXiv:1803.03324. Retrieved from https://arxiv.org/abs/1803.03324
[106] Chenghua Lin, Yulan He, Richard Everson, and Stefan Ruger. 2012. Weakly supervised joint sentiment-topic detection from text. *IEEE Transactions on Knowledge and Data Engineering* 24, 6 (June 2012), 1134–1145.
[107] Mingjie Ling, Qiaohong Chen, Qi Sun, and Yubo Jia. 2020. Hybrid neural network for sina weibo sentiment analysis. *IEEE Transactions on Computational Social Systems* 7, 4 (Aug. 2020), 983–990.
[108] Bing Liu. 2015. *Sentiment Analysis: Mining Opinions, Sentiments, and Emotions*. Cambridge University Press.
[109] Ruijun Liu, Yuqian Shi, Changjiang Ji, and Ming Jia. 2019. A survey of sentiment analysis based on transfer learning. *IEEE Access* 7 (2019), 85401–85412. DOI: https://doi.org/10.1109/ACCESS.2019.2925059
[110] Xia Liu, Alvin C. Burns, and Yingjian Hou. 2017. An investigation of brand-related user-generated content on Twitter. *Journal of Advertising (Routledge* 46, 2 (2017), 236–247. DOI: https://doi.org/10.1080/00913367.2017.1297273
[111] Zhifeng Liu, Tingting Zhang, and Qiujun Lan. 2014. An extended SISa model for sentiment contagion. *Discrete Dynamics in Nature and Society (Hindawi)* (2014), 1–7. DOI: https://doi.org/10.1155/2014/262384
[112] Qiang Lu, Xia Sun, Yunfei Long, Zhizezhang Gao, Jun Feng, and Tao Sun. 2024. Sentiment analysis: Comprehensive reviews, recent advances, and open challenges. *IEEE Transactions on Neural Networks and Learning Systems* 35, 11 (2024), 15092–15112. DOI: https://doi.org/10.1109/TNNLS.2023.3294810
[113] Christopher D. Manning, Mihai Surdeanu, John Bauer, Jenny Finkel, Steven J. Bethard, and David McClosky. 2014. The stanford CoreNLP natural language processing toolkit. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics: System Demonstrations*. Association for Computational Linguistics, 55–60.
[114] Zhifei Mao, Yuming Jiang, Geyong Min, Supeng Leng, Xiaolong Jin, and Kun Yang. 2017. Mobile social networks: Design requirements, architecture, and state-of-the-art technology. *Computer Communications* 100 (Mar 2017), 1–19.
[115] Tahir Maqsood, Osman Khalid, Rizwana Irfan, Sajjad A. Madani, and Samee U. Khan. 2016. Scalability issues in online social networks. *ACM Computing Surveys* 49, 2, Article 40 (Sep 2016), 42 pages. DOI: https://doi.org/10.1145/2968216
[116] André C. R. Martins. 2008. Continuous opinions and discrete actions in opinion dynamics problems. *International Journal of Modern Physics C* 19, 04 (Apr 2008), 617–624. DOI: https://doi.org/10.1142/S0129183108012339
[117] W. McDougall. 1912. The principal instincts and the primary emotions of man. In *An Introduction to Social Psychology (4th ed., rev. With Supplementary Chapter on Theories of Action*. John W. Luce & Company, 45–89.
[118] A. Mehrabian. 1996. Pleasure-arousal-dominance: A general framework for describing and measuring individual differences in Temperament. *Current Psychology* 14, 4 (1996), 261–292.
[119] Ethan Mendes, Yang Chen, Wei Xu, and Alan Ritter. 2023. Human-in-the-loop evaluation for early misinformation detection: A case study of COVID-19 treatments. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. ACL, Toronto, Canada, 15817–15835.
[120] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S. Corrado, and Jeff Dean. 2013. Distributed representations of words and phrases and their compositionality. In *Proceedings of the Advances in Neural Information Processing Systems*, C. J. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K.Q. Weinberger (Eds.), Vol. 26. Curran Associates, Inc.

[121] S. Milgram. 1967. The small world problem. *Psychology Today* 2, 1 (1967), 60–67. DOI: https://doi.org/10.1068/a090185
[122] Saif Mohammad. 2018. Obtaining reliable human ratings of valence, arousal, and dominance for 20,000 English words. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers).* 174–184.
[123] Nasir Naveed, Thomas Gottron, Jérôme Kunegis, and Arifah Che Alhadi. 2011. Bad news travel fast: A content-based analysis of interestingness on Twitter. In *Proceedings of the 3rd International Conference on Web Science.* ACM 1–7.
[124] M. E. J. Newman. 2002. Assortative mixing in networks. *Physical Review Letters* 89, 20 (Nov. 2002), 208701.
[125] M. E. J. Newman. 2005. A measure of betweenness centrality based on random walks. *Social Networks* 27, 1 (Jan. 2005), 39–54. DOI: https://doi.org/10.1016/j.socnet.2004.11.009 WOS:000227500500003.
[126] John R. Nofsinger. 2003. Social mood and financial economics. *The Journal of Behavioral Finance* 6, 3 (2003), 129–143.
[127] Fatemeh Noroozi, Ciprian Adrian Corneanu, Dorota Kamińska, Tomasz Sapiński, Sergio Escalera, and Gholamreza Anbarjafari. 2021. Survey on emotional body gesture recognition. *IEEE Transactions on Affective Computing* 12, 2 (2021), 505–523. DOI: https://doi.org/10.1109/TAFFC.2018.2874986
[128] Adenrele Oduntan, Oladapo Oyebode, Amelia Hernandez Beltran, Jonathon Fowles, Darren Steeves, and Rita Orji. 2022. "I Let Depression and Anxiety Drown Me...": Identifying factors associated with resilience based on journaling using machine learning and thematic analysis. *IEEE Journal of Biomedical and Health Informatics* 26, 7 (2022), 3397–3408. DOI: https://doi.org/10.1109/JBHI.2022.3149862
[129] OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, et al. 2024. GPT-4 Technical Report. arXiv:2303.08774. Retrieved from https://arxiv.org/abs/2303.08774
[130] A. Ortony and T. J. Turner. 1990. What's basic about basic emotions. *Psychology Review* 97, 3 (1990), 315–331.
[131] P. Erdős and A. Rényi. 1959. On random graphs I. *Publicationes Mathematicae* 6, 3-4 (1959), 290–297.
[132] Georgios Paltoglou. 2016. Sentiment-based event detection in Twitter. *Journal of the Association for Information Science and Technology (Wiley)* 67, 7 (2016), 1576–1587. DOI: https://doi.org/10.1002/asi.23465
[133] J. Panksepp. 2005. On the embodied neural nature of core emotional affects. *Journal of Consciousness Studies* 12, 8-10 (2005), 158–184.
[134] Michael J. Paul and Mark Dredze. 2011. You are what you tweet: Analyzing Twitter for public health. In *Proceedings of the 5th International AAAI Conference on Weblogs and Social Media.* 265–272. DOI: https://doi.org/10.1.1.224.9974
[135] David Peleg. 2002. Local majorities, coalitions and monopolies in graphs: A review. *Theoretical Computer Science* 282, 2 (Jun 2002), 231–257. DOI: https://doi.org/10.1016/S0304-3975(01)00055-X
[136] Jeffrey Pennington, Richard Socher, and Christopher Manning. 2014. GloVe: Global vectors for word representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP'14).* ACL, Doha, Qatar, 1532–1543. DOI: https://doi.org/10.3115/v1/D14-1162
[137] Matthew E. Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers).* ACL, New Orleans, Louisiana, 2227–2237. DOI: https://doi.org/10.18653/v1/N18-1202
[138] Trinh Pham, Khoi Le, and Anh Tuan Luu. 2024. UniBridge: A unified approach to cross-lingual transfer learning for low-resource languages. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers).* ACL, Bangkok, Thailand, 3168–3184. https://aclanthology.org/2024.acl-long.174
[139] Robert Plutchik. 2001. The nature of emotions. *American Scientist* 89, 4 (2001), 344–350.
[140] Jonathan Posner, James A. Russell, and Bradley S. Peterson. 2005. The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. *Development and Psychopathology* 17, 03 (2005), 3279–3288. DOI: https://doi.org/10.1017/S0954579405050340
[141] Dian Purworini and Rona Rizkhy Bunga Chasana. 2024. The COVID-19 crisis response strategy and public opinion: The complexities involved in acquiring public support for the sticky crisis. *Journal of Contemporary Eastern Asia* 23, 2 (12 2024), 18–34.
[142] Yujia Qin, Fanchao Qi, Sicong Ouyang, Zhiyuan Liu, Cheng Yang, Yasheng Wang, Qun Liu, and Maosong Sun. 2020. Improving sequence modeling ability of recurrent neural networks via sememes. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 28 (2020), 2364–2373. DOI: https://doi.org/10.1109/TASLP.2020.3012060
[143] F. Radicchi, C. Castellano, F. Cecconi, V. Loreto, and D. Parisi. 2004. Defining and identifying communities in networks. *Proceedings of the National Academy of Sciences of the United States of America* 101, 9 (March 2004), 2658–2663.
[144] Biswarup Ray, Avishek Garain, and Ram Sarkar. 2021. An ensemble-based hotel recommender system using sentiment analysis and aspect categorization of hotel reviews. *Applied Soft Computing* 98 (2021), 106935.
[145] Rui Ren, Desheng Dash Wu, and Tianxiang Liu. 2019. Forecasting stock market movement direction using sentiment analysis and support vector machine. *IEEE Systems Journal* 13, 1 (March 2019), 760–770.

[146] Ramon Gouveia Rodrigues, Rafael Marques das Dores, Celso G. Camilo-Junior, and Thierson Couto Rosa. 2016. Sentihealth-cancer: A sentiment analysis tool to help detecting mood of patients in online social networks. *International Journal of Medical Informatics (Elsevier)* 85, 1 (2016), 80–95. DOI: https://doi.org/10.1016/j.ijmedinf.2015.09.007
[147] Everett M. Rogers. 2013. *Diffusion of Innovations*. Simon and Schuster.
[148] James A. Russell. 1980. A circumplex model of affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161–1178.
[149] James A. Russell. 1991. Culture and the categorization of emotions. *Psychological Bulletin* 110, 3 (1991), 426–450.
[150] Hamann S. 2012. Mapping discrete and dimensional emotions onto the brain: Controversies and consensus. *Trends in Cognitive Sciences* 9, 16 (2012), 458–466.
[151] Evangelos Sariyanidi, Hatice Gunes, and Andrea Cavallaro. 2015. Automatic analysis of facial affect: A survey of registration, representation, and recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 37, 6 (2015), 1113–1133.
[152] Klaus R. Scherer. 2005. What are emotions? And how can they be measured? *Social Science Information* 44, 4 (2005), 695–729.
[153] Klaus R. Scherer, Angela Schorr, and Tom Johnstone. 2001. *Appraisal Processes in Emotion: Theory, Methods, Research*. Oxford University Press.
[154] Harold Schlosberg. 1954. Three dimensions of emotion. *Psychological Review* 61, 2 (1954), 81.
[155] Thanveer Shaik, Xiaohui Tao, Christopher Dann, Haoran Xie, Yan Li, and Linda Galligan. 2023. Sentiment analysis and opinion mining on educational data: A survey. *Natural Language Processing Journal* 2 (2023), 100003.
[156] Paulo Shakarian, Abhinav Bhatnagar, Ashkan Aleali, Elham Shaabani, and Ruocheng Guo. 2015. *The Independent Cascade and Linear Threshold Models*. Springer International Publishing, Cham, 35–48.
[157] Xia-Meng Si, Wen-Dong Wang, and Yan Ma. 2016. Role of propagation thresholds in sentiment-based model of opinion evolution with information diffusion. *Physica A (Elsevier)* 451 (2016), 549–559.
[158] Stefan Siersdorfer, Sergiu Chelaru, Wolfgang Nejdl, and Jose San Pedro. 2010. How useful are your comments? Analyzing and predicting youtube comments and comment ratings. In *Proceedings of the 19th International Conference on World Wide Web*. ACM, New York, NY, USA, 891–900. DOI: https://doi.org/10.1145/1772690.1772781
[159] Antonios Siganos, Evangelos Vagenas-Nanos, and Patrick Verwijmeren. 2014. Facebook'S daily sentiment and international stock markets. *Journal of Economic Behavior and Organization (Elsevier)* 107 (2014), 730–743.
[160] Oskar Skibski, Tomasz P. Michalak, and Talal Rahwan. 2018. Axiomatic characterization of game-theoretic centrality. *Journal of Artificial Intelligence Research* 62, 1 (May 2018), 33–68. DOI: https://doi.org/10.1613/jair.1.11202
[161] Zhijie Song, Rui Shi, Jie Jia, and Jian Wang. 2016. Sentiment contagion based on the modified SOSa-SPSa model. *Computational and Mathematical Methods in Medicine* (2016), 1–7. DOI: https://doi.org/10.1155/2016/9682538
[162] Stefan Stieglitz and Linh Dang-Xuan. 2012. Political communication and influence through microblogging–an empirical analysis of sentiment in Twitter messages and retweet behavior. In *Proceedings of the 2012 45th Hawaii International Conference on System Sciences*. 3500–3509. DOI: https://doi.org/10.1109/HICSS.2012.476
[163] Teng Sun, Liqiang Jing, Yinwei Wei, Xuemeng Song, Zhiyong Cheng, and Liqiang Nie. 2023. Dual consistency-enhanced semi-supervised sentiment analysis towards COVID-19 tweets. *IEEE Transactions on Knowledge and Data Engineering* 35, 12 (2023), 12605–12617. DOI: https://doi.org/10.1109/TKDE.2023.3270940
[164] Xia Sun, Yi Gao, Richard Sutcliffe, Shou-Xi Guo, Xin Wang, and Jun Feng. 2021. Word representation learning based on bidirectional GRUs with drop loss for sentiment classification. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 51, 7 (July 2021), 4532–4542. DOI: https://doi.org/10.1109/TSMC.2019.2940097
[165] Anjana Susarla, Jeong-Ha Oh, and Yong Tan. 2012. Social networks and the diffusion of user-generated content: Evidence from YouTube. *Information Systems Research* 23, 1 (Mar 2012), 23–41. DOI: https://doi.org/10.1287/isre.1100.0339
[166] Maite Taboada, Julian Brooke, Milan Tofiloski, Kimberly Voll, and Manfred Stede. 2011. Lexicon-based methods for sentiment analysis. *Computational Linguistics* 37, 2 (June 2011), 267–307. DOI: https://doi.org/10.1162/COLI_a_00049
[167] Jie Tang and Acm Fong. 2013. Sentiment diffusion in large scale social networks. In *Proceedings of the 2013 IEEE International Conference on Consumer Electronics (ICCE'13)*. 244–245. DOI: https://doi.org/10.1109/ICCE.2013.6486878
[168] Mani Kumar Tellamekala, Shahin Amiriparian, Björn W. Schuller, Elisabeth André, Timo Giesbrecht, and Michel Valstar. 2024. COLD fusion: Calibrated and ordinal latent distribution fusion for uncertainty-aware multimodal emotion recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 2 (2024), 805–822.
[169] Hao Tian, Can Gao, Xinyan Xiao, Hao Liu, Bolei He, Hua Wu, Haifeng Wang, and Feng Wu. 2020. SKEP: Sentiment knowledge enhanced pre-training for sentiment analysis. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. ACL, Online, 4067–4076. DOI: https://doi.org/10.18653/v1/2020.acl-main.374
[170] Silvan S. Tomkins. 2008. *Affect Imagery Consciousness: The Complete Edition: Two Volumes*. Springer publishing company.

[171] Mikalai Tsytsarau, Themis Palpanas, and Malu Castellanos. 2014. Dynamics of news events and social media reaction. In *Proceedings of the 20th ACM SIGKDD*. ACM, 901–910. DOI: https://doi.org/10.1145/2623330.2623670

[172] E. B. Varghese and S. M. Thampi. 2018. A deep learning approach to predict crowd behavior based on emotion. In *Proceedings of the International Conference on Smart Multimedia* (2018), 296–307.

[173] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems* (Long Beach, California, USA) *(NIPS'17)*. Curran Associates Inc., Red Hook, NY, USA, 6000–6010.

[174] Feng Wang, Guojun Wang, and Dongqing Xie. 2016. Maximizing the spread of positive influence under LT-MLA model. In *Proceedings of the The 10th Asia-Pacific Services Computing Conference (APSCC, also in Springer LNCS 10065)*, Vol. 2. 450–463.

[175] Huanhuan Wang, Qun Yang, Lei Fang, and Weihua Lei. 2015. Maximizing positive influence in signed social networks. In *Proceedings of the International Conference on Cloud Computing and Security (Springer LNCS 9483)*. 356–367.

[176] Jin Wang, Liang-Chih Yu, K. Robert Lai, and Xuejie Zhang. 2020. Tree-structured regional CNN-LSTM model for dimensional sentiment analysis. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 28 (2020), 581–591. DOI: https://doi.org/10.1109/TASLP.2019.2959251

[177] Qiyao Wang, Yuehui Jin, Tan Yang, and Shiduan Cheng. 2017. An emotion-based independent cascade model for sentiment spreading. *Knowledge-Based Systems* 116 (2017), 86–93. DOI: https://doi.org/10.1016/j.knosys.2016.10.029

[178] Qiyao Wang, Zhen Lin, Yuehui Jin, Shiduan Cheng, and Tan Yang. 2015. ESIS: Emotion-based spreader-ignorant-stifler model for information diffusion. *Knowledge-Based Systems (Elsevier)* 81 (2015), 46–55.

[179] Ranran Wang, Yin Zhang, Wenchao Wan, Xiong Li, and Min Chen. 2024. Predicting multi-scale information diffusion via minimal substitution neural networks. In *Proceedings of the IEEE Conference on Computer Communications (INFOCOM'24)*. 1890–1899.

[180] Zhishuang Wang, Chengyi Xia, Zengqiang Chen, and Guanrong Chen. 2021. Epidemic propagation with positive and negative preventive information in multiplex networks. *IEEE Transactions on Cybernetics* 51, 3 (2021), 1454–1462.

[181] D. Waston and L. A. Clark. 1997. Measurement and mismeasurement of mood: Recurrent and emergent issues. *Journal of Personality Assessment* 68, 2 (1997), 267–296.

[182] David Watson and Auke Tellegen. 1985. Toward a consensual structure of mood. *Psychonological Bulletin* 98, 2 (1985), 219–235. DOI: https://doi.org/10.1037/0033-2909.98.2.219

[183] Duncan J. Watts and Steven H. Strogatz. 1998. Collective dynamics of 'small-world' networks. *Nature* 393, 6684 (June 1998), 440–442. DOI: https://doi.org/10.1038/30918

[184] Sheng Wen, Mohammad Sayad Haghighi, Chao Chen, Yang Xiang, Wanlei Zhou, and Weijia Jia. 2015. A sword with two edges: Propagation studies on both positive and negative information in online social networks. *IEEE Transactions on Computers* 64, 3 (Mar 2015), 640–653. DOI: https://doi.org/10.1109/TC.2013.2295802

[185] Cynthia M. Whissell. 1989. The dictionary of affect in language. In *Proceedings of the Measurement of Emotions*. Academic Press, 113–131. DOI: https://doi.org/10.1016/B978-0-12-558704-4.50011-6

[186] Theresa Wilson, Janyce Wiebe, and Paul Hoffmann. 2005. Recognizing contextual polarity in phrase-level sentiment analysis. In *Proceedings of the Conference on Human Language Technology and Empirical Methods in Natural Language Processing* (Vancouver, British Columbia, Canada) *(HLT'05)*. ACL, USA, 347–354. DOI: https://doi.org/10.3115/1220575.1220619

[187] Desheng Dash Wu, Lijuan Zheng, and David L. Olson. 2014. A decision support approach for online stock forum sentiment analysis. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 44, 8 (Aug. 2014), 1077–1087.

[188] Wilhelm Max Wundt and Charles Hubbard Judd. 1897. *Outlines of Psychology (Vol.1)*. Scholarly Press.

[189] Rui Xia, Jie Jiang, and Huihui He. 2017. Distantly supervised lifelong learning for large-scale social media sentiment analysis. *IEEE Transactions on Affective Computing* 8, 4 (2017), 480–491. DOI: https://doi.org/10.1109/TAFFC.2017.2771234

[190] Xi Xiong, Yuanyuan Li, Shaojie Qiao, Nan Han, Yue Wu, Jing Peng, and Binyong Li. 2018. An emotional contagion model for heterogeneous social media with multiple behaviors. *Physica A* 490 (2018), 185–202.

[191] Peng Xu, Andrea Madotto, Chien-Sheng Wu, Ji Ho Park, and Pascale Fung. 2018. Emo2Vec: Learning generalized emotion representation by multi-task training. In *Proceedings of the 9th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*. ACL, 292–298. DOI: https://doi.org/10.18653/v1/W18-6243

[192] Ali Yadollahi, Ameneh Gholipour Shahraki, and Osmar R. Zaiane. 2017. Current state of text sentiment analysis from opinion to emotion mining. *Comput. Surveys* 50, 2 (May 2017), 1–33. DOI: https://doi.org/10.1145/3057270

[193] Liang Yang, Xiaochun Cao, Dongxiao He, Chuan Wang, Xiao Wang, and Weixiong Zhang. 2016. Modularity based community detection with deep learning. In *Proceedings of the 25th International Joint Conference on Artificial Intelligence* (New York, New York, USA) *(IJCAI'16)*. AAAI Press, 2252–2258.

[194] Zi Yang, Jie Tang, Juanzi Li, and Wenjun Yang. 2011. Social community analysis via a factor graph model. *IEEE Intelligent Systems* 26, 3 (2011), 58–65. DOI: https://doi.org/10.1109/MIS.2010.55

[195] Shuo Yin and Guoqiang Zhong. 2024. TextGT: A double-view graph transformer on text for aspect-based sentiment analysis. *Proceedings of the AAAI Conference on Artificial Intelligence* 38, 17 (2024), 19404–19412.

[196] Jiaxuan You, Rex Ying, Xiang Ren, William L. Hamilton, and Jure Leskovec. 2018. GraphRNN: Generating realistic graphs with deep auto-regressive models. In *Proceedings of the International Conference on Machine Learning (ICML'18)*.

[197] Huiyuan Zhang, Thang N. Dinh, and My T. Thai. 2013. Maximizing the spread of positive influence in online social networks. In *Proceedings of the IEEE International Conference on Distributed Computing Systems (ICDCS'13)*. 317–326.

[198] Haizheng Zhang, Baojun Qiu, C. Lee Giles, Henry C. Foley, and John Yen. 2007. An LDA-based community structure discovery approach for large-scale social networks. In *Proceedings of the 2007 IEEE Intelligence and Security Informatics*. 200–207.

[199] Taolin Zhang, Ruyao Xu, Chengyu Wang, Zhongjie Duan, Cen Chen, Minghui Qiu, Dawei Cheng, Xiaofeng He, and Weining Qian. 2023. Learning knowledge-enhanced contextual language representations for domain natural language understanding. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. ACL, Singapore, 15663–15676. DOI: https://doi.org/10.18653/v1/2023.emnlp-main.969

[200] Wenxuan Zhang, Yue Deng, Bing Liu, Sinno Pan, and Lidong Bing. 2024. Sentiment analysis in the era of large language models: A reality check. In *Proceedings of the Findings of the Association for Computational Linguistics: NAACL 2024*, Kevin Duh, Helena Gomez, and Steven Bethard (Eds.). Association for Computational Linguistics, Mexico City, Mexico, 3881–3906. DOI: https://doi.org/10.18653/v1/2024.findings-naacl.246

[201] Laijun Zhao, Jiajia Wang, Rongbing Huang, Hongxin Cui, Xiaoyan Qiu, and Xiaoli Wang. 2014. Sentiment contagion in complex networks. *Physica A (Elsevier)* 394 (2014), 17–23. DOI: https://doi.org/10.1016/j.physa.2013.09.057

[202] Wei Zhao, Ziyu Guan, Long Chen, Xiaofei He, Deng Cai, Beidou Wang, and Quan Wang. 2018. Weakly-supervised deep embedding for product review sentiment analysis. *IEEE Transactions on Knowledge and Data Engineering* 30, 1 (Jan. 2018), 185–197. DOI: https://doi.org/10.1109/TKDE.2017.2756658

[203] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. 2025. A survey of large language models. arXiv:2303.18223. Retrieved from https://arxiv.org/abs/2303.18223

[204] Yuxin Zhao, Sixiang Cheng, Xiaoyan Yu, and Huilan Xu. 2020. Chinese public's attention to the COVID-19 epidemic on social media: Observational descriptive study. *Journal of Medical Internet Research* 22, 5 (2020), 1–13.

[205] Zaixiang Zheng, Shujian Huang, Rongxiang Weng, Xin-Yu Dai, and Jiajun Chen. 2020. Improving self-attention networks with sequential relations. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 28 (2020), 1707–1716. DOI: https://doi.org/10.1109/TASLP.2020.2996807

[206] Changtao Zhong and Nishanth Sastry. 2017. Systems applications of social networks. *ACM Computing Surveys* 50, 5, Article 63 (Sep 2017), 42 pages. DOI: https://doi.org/10.1145/3092742

[207] Huan Zhou, Linping Tong, Shouzhi Xu, Chungming Huang, and Jialu Fan. 2016. Predicting temporal centrality in opportunistic mobile social networks based on social behavior of people. *Personal and Ubiquitous Computing* 20, 6 (Nov. 2016), 885–897. DOI: https://doi.org/10.1007/s00779-016-0958-0

[208] Jin Zhou and Jun-min Ye. 2023. Sentiment analysis in education research: A review of journal publications. *Interactive Learning Environments* 31, 3 (2023), 1252–1264.

[209] Xiaokang Zhou, Wei Liang, Zijia Luo, and Yi Pan. 2021. Periodic-aware intelligent prediction model for information diffusion in social networks. *IEEE Transactions on Network Science and Engineering* 8, 2 (2021), 894–904.

[210] Fuzhen Zhuang, Zhiyuan Qi, Keyu Duan, Dongbo Xi, Yongchun Zhu, Hengshu Zhu, Hui Xiong, and Qing He. 2021. A comprehensive survey on transfer learning. *Proceedings of the IEEE* 109, 1 (2021), 43–76.

Received 4 December 2021; revised 6 February 2025; accepted 18 April 2025